



Class-10th
Chapter-05

Chemical Bonding

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Octet Rule, limitations of octet rule, The Lewis Structures, Multiple Covalent Bonds, Ionic or Electrovalent Bonds, Formal charge

- The sulphur compound in which the sulphur atom has octet configuration in its valence shell, among the following is
 (A) Sulphur trioxide
 (B) Sulphur hexafluoride
 (C) Sulphur dichloride
 (D) Sulphur dioxide
- In which one of the following compounds does the central atom obey the octet rule?
 (A) XeF_4 (B) XeOF_2
 (C) SCl_2 (D) AlCl_3
- The attractive force which holds various constituents together in different chemical species is called
 (A) valence bond
 (B) chemical bond
 (C) atomic bond
 (D) electrovalent bond
- The octet rule is observed in
 (A) PCl_5 (B) CO_2
 (C) BCl_3 (D) SF_6
- Which of the following is electron-deficient?
 (A) $(\text{BH}_3)_2$ (B) PH_3
 (C) $(\text{CH}_3)_2$ (D) $(\text{SiH}_3)_2$
- The nature of bonding between adjacent atoms in graphite is
 (A) covalent
 (B) ionic
 (C) van der Waals
 (D) metallic
- Which one of the following contains ionic, covalent and co-ordinate bonds?
 (A) NaOH (B) NaCl
 (C) HCl (D) NaN_3
- Which of the following is not a characteristic of a covalent compound?
 (A) Low melting point
 (B) No definite geometry
 (C) Insoluble in polar solvent
 (D) Small difference in electronegativity between the combining atoms.
- The electronegativity of an element is low. The bond formed between two identical atoms of the above element is most likely to be
 (A) covalent
 (B) metallic
 (C) ionic
 (D) co-ordinate covalent
- The type of bonds present in $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ _____.
 (A) electrovalent and covalent
 (B) electrovalent and coordinate
 (C) electrovalent, covalent and coordinate
 (D) covalent and coordinate
- Which of the following is correct for the process of formation of a chemical bond?
 (A) Electron-electron repulsion becomes more than the nucleus-electron attraction.
 (B) Energy of the system does not change.
 (C) Energy increases.
 (D) Energy decreases.
- The compound containing coordinate bond is
 (A) SO_3 (B) O_3
 (C) H_2SO_2 (D) all of these
- If stability were attained with 6 electrons rather than with 8 electrons, what would be the formula of the stable fluoride ion
 (A) F^{3+} (B) F^+
 (C) F^- (D) F^{2-}
- Duplet configuration is not found in
 (A) Hydride ion
 (B) Hydrogen molecule
 (C) Lithium cation
 (D) Be^{3+}

15. The element having highest valency with respect to oxygen is
 (A) Sodium
 (B) Aluminium
 (C) Chlorine
 (D) Sulphur
16. Polarisation is the distortion of the shape of anion by an adjacently placed cation. Which of the following statement is correct?
 (A) Maximum polarization is brought about by cation of high charge
 (B) Maximum polarization is brought about by cation of Low charge
 (C) A large cation is likely to bring about a large degree of polarization
 (D) A small anion is likely undergo a large degree of polarization
17. Example of super octet molecule is:
 (A) ClF_3 (B) PCl_5
 (C) IF_7 (D) All
18. Molecule having an incomplete octet in the central atom is
 (A) PCl_5 (B) NH_3
 (C) BCl_3 (D) H_2O
19. In PO_4^{3-} , the formal charge on each oxygen atom and the P -O bond order respectively are
 (A) -0.75, 0.6 (B) -0.75, 1.0
 (C) -0.75, 1.25 (D) -3, 1.25
20. The correct order of decreasing polarity is
 (A) $\text{HF} > \text{SO}_2 > \text{H}_2\text{O} > \text{NH}_3$
 (B) $\text{HF} > \text{H}_2\text{O} > \text{SO}_2 > \text{NH}_3$
 (C) $\text{HF} > \text{NH}_3 > \text{SO}_2 > \text{H}_2\text{O}$
 (D) $\text{H}_2\text{O} > \text{NH}_3 > \text{SO}_2 > \text{HF}$
21. In which of the following molecules octet rule is not followed?
 (A) NH_3 (B) CH_4
 (C) CO_2 (D) NO
22. Which of the following shows the Lewis dot formula for CO_2 ?
 (A) $:\ddot{\text{O}}::\text{C}::\ddot{\text{O}}:$ (B) $:\ddot{\text{O}}:\ddot{\text{C}}::\ddot{\text{O}}:$
 (C) $:\ddot{\text{O}}:\text{C}::\ddot{\text{O}}:$ (D) $:\ddot{\text{O}}:\ddot{\text{C}}::\ddot{\text{O}}:$
23. In which of the following species the bond is nondirectional?
 (A) NCl_3 (B) RbCl
 (C) BeCl_2 (D) BCl_3
24. Sodium chloride has a crystalline structure made up of Na^+ and Cl^- ions. Why does NaCl not conduct electricity in solid state?
 (A) The ions of NaCl become mobile only in molten state and are not free to move in solid state
 (B) The crystalline structure does not have ions
 (C) When a bond is formed between ions they lose their charges
 (D) None of these
25. Most ionic compounds have
 (a) high melting point and low boiling point
 (b) high melting point and non-directional bonds
 (c) high solubilities in polar solvents and low solubilities in non-polar solvents
 (d) three dimensional network structures and are good conductors of electricity in the molten state
 (A) None of the above
 (B) b,c,d
 (C) a,b,c
 (D) All of the above
26. Which of the following does not conduct electricity.
 (A) Molten NaOH
 (B) Molten KOH
 (C) Solid NaCl
 (D) Aqueous NaCl

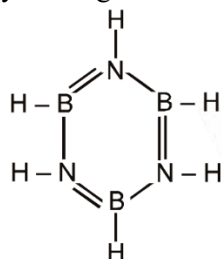
27. In an ionic compound A^+X^- the degree of covalent bonding is greatest when:
 (A) A^+ and X^- ion are small
 (B) A^+ and X^- ion are large
 (C) A^+ and X^- ion are approximately of the same size
 (D) A^+ is small and X^- is large
28. The most covalent halide of s-Block is
 (A) CsF (B) CsBr
 (C) BeI₂ (D) BeF₂
29. The maximum covalency of representative elements is equal to (excluding 1st and 2nd period)
 (A) the number of unpaired p-electrons
 (B) the number of paired d-electrons
 (C) the number of paired s and p -electrons
 (D) the actual number of s and p -electrons in the outermost shell
30. Which of the following contains both electrovalent and covalent bonds?
 (A) MgCl₂ (B) H₂O
 (C) NH₄Cl (D) None
31. The types of bond present in N₂O₅ are
 (A) only covalent
 (B) only ionic
 (C) ionic and covalent
 (D) covalent and coordinate
32. The octet rule is not obeyed in
 (A) CO₂ (B) BCl₃
 (C) PCl₅ (D) (B) and (C) both
33. For the formation of covalent bond the difference in the value of electronegativity should be
 (A) 1.7
 (B) More than 1.7
 (C) 1.7 or more
 (D) equal to or less than 1.7
34. NH₃ and BF₃ combine readily because of the formation of
 (A) a covalent bond (B) a hydrogen bond
 (C) a coordinate bond (D) an ionic bond
35. The covalency of nitrogen in HNO₃ is
 (A) 0 (B) 3
 (C) 4 (D) 5
36. Which one of the following molecules has a coordinate as well as covalent bond?
 (A) NH₄Cl (B) AlCl₃
 (C) NaCl (D) Cl₂
37. Which of the following species are hypervalent?
 1. ClO₄⁻ 2. BF₃
 3. SO₄²⁻ 4. CO₃²⁻
 (A) 1,2,3 (B) 1,3
 (C) 3,4 (D) 1,2
38. If the atomic number of element X is 7 the Lewis diagram for the element is
 (A) X · (B) · $\ddot{X}$:
 (C) · $\ddot{X}$: (D) : $\ddot{X}$:
39. Which of the following Lewis diagram is incorrect?
 (A) Na⁺ [: $\ddot{O}$: - $\ddot{Cl}$:]⁻ (B) $\begin{array}{c} :\ddot{Cl}: \\ | \\ :\ddot{C}: - \ddot{C}: \\ | \\ :\ddot{Cl}: \end{array}$
 (C) : $\ddot{O}$:C::: $\ddot{O}$: (D) $\begin{array}{c} :\ddot{Cl}: \\ | \\ H - \ddot{N} - \ddot{N} - H \\ | \quad | \\ H \quad H \end{array}$
40. What are the formal charges on central sulphur and each terminal oxygen atoms in SO₂?
 (A) 0,0,0 (B) +2,0,-1
 (C) 0,-1,+1 (D) +2,+2,+2
41. Which of the following compounds does not follow the octet rule for electron distribution
 (A) PCl₅ (B) PCl₃
 (C) H₂O (D) PH₃

42. The valency of A=3 and B=2, then the compound is
 (A) A_2B_3 (B) A_3B_2
 (C) A_3B_3 (D) A_2B_2

43. Which of the following does not represent the correct Lewis structure for the compounds given?

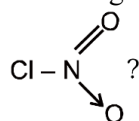
- (A) $MgO - [Mg]^{2+} [:\ddot{O}:]^{2-}$
 (B) $CN^- - :C:::N:$
 (C) BrF_3 (Br in center) $\begin{array}{c} :\ddot{F}: - \ddot{Br} - :\ddot{F}: \\ | \\ :\ddot{F}: \end{array}$
 (D) All of these

44. What are the formal charges on B and N respectively in the given structure?



- (A) -1, +1 (B) +1, -1
 (C) -2, +2 (D) 0, 0

45. What is the formal charge on N in



- (A) +1 (B) -1
 (C) 0 (D) +2

46. The pair of compounds which can form a coordinate bond is:

- (A) $(C_2H_5)_3B$ and $(CH_3)_3N$
 (B) HCl and HBr
 (C) BF_3 and NH_3
 (D) (A) and (C) both

47. In which compound coordinate bond is present:

- (A) NH_3 (B) NH_4OH
 (C) H_2O (D) HCl

48. Among $LiCl$, $BeCl_2$, BCl_3 and CCl_4 , the covalent bond character follows the order-

- (A) $LiCl < BeCl_2 > BCl_3 > CCl_4$
 (B) $LiCl > BeCl_2 < BCl_3 < CCl_4$
 (C) $LiCl < BeCl_2 < BCl_3 < CCl_4$
 (D) $LiCl > BeCl_2 > BCl_3 > CCl_4$

Ionic Bond: Factors which favour the formation of ionic bond, Lattice Enthalpy, Hydration Energy, Solubility of Ionic Salt, Thermal Stability

49. CaO and $NaCl$ have the same crystal structure and approximately the same ionic radii. If U is the lattice energy of $NaCl$, the approximate lattice energy of CaO is

- (A) $U/2$ (B) U
 (C) $2U$ (D) $4U$

50. During bond formation, potential energy of the system

- (A) Increases
 (B) Decreases
 (C) Remains the same
 (D) Cannot be predicted

51. Which of the following is not an ionic compound

- (A) CaC_2 (B) NaH
 (C) BaF_2 (D) BF_3

52. Which of the following is not a correct statement about an ionic compound

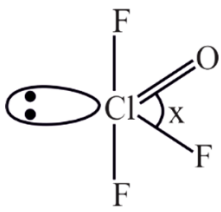
- (A) The higher the temperature, the more the solubility.
 (B) The higher the dielectric constant of the solvent, the more the solubility.
 (C) The higher the dipole moment of the solvent, the more the solubility.
 (D) The higher the lattice energy, the more the solubility.

53. Lattice energy depends on

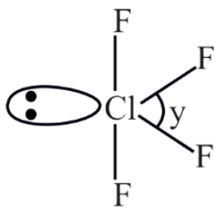
- (A) Only radius of cation
 (B) Only radius of anion
 (C) Cation to anion radius ratio
 (D) Sum of the radii of cation and anion

54. Number of electrons transferred from one atom to another during bond formation in SrS (Strontium Sulphide)
 (A) 1 (B) 2
 (C) 3 (D) 4
55. Which of the following has the lowest Lattice energy?
 (A) LiF (B) LiCl
 (C) LiBr (D) LiI
56. Which one of the following has the highest lattice energy?
 (A) LiCl (B) CaCl₂
 (C) LiF (D) KCl
57. Amongst LiCl, RbCl, BeCl₂ and MgCl₂, the compounds with the greatest and the least ionic character respectively are:
 (A) LiCl and RbCl
 (B) RbCl and BeCl₂
 (C) RbCl and MgCl₂
 (D) MgCl₂ and BeCl₂
58. Lattice energy of a solid increases if
 (A) The radii of ions are small
 (B) The ions are neutral
 (C) Charges on the ions are small
 (D) Madelung constant decreases
59. The decreasing order of ionic nature of the following compound is:
 (A) LiI > NaBr > KCl > CsF
 (B) LiI > KCl > NaBr > CsF
 (C) CsF > NaBr > KCl > LiI
 (D) CsF > KCl > NaBr > LiI
60. Correct order of melting point (°C) of the following compound: NaBr, MgBr, AlBr₃
 (A) 91.5, 700, 755
 (B) 755, 700, 91.5
 (C) 700, 755, 91.5
 (D) 755, 91.5, 700
61. The correct order of polarizability of X⁻ is:
 (A) F⁻ > Cl⁻ > Br⁻ > I⁻
 (B) Cl⁻ > F⁻ > Br⁻ > I⁻
 (C) Br⁻ > Cl⁻ > F⁻ > I⁻
 (D) I⁻ > Br⁻ > Cl⁻ > F⁻
62. The ionic compounds will be sparingly soluble if.
 (A) Hydration energy = lattice energy
 (B) Hydration energy > lattice energy
 (C) Hydration energy < lattice energy
 (D) Hydration energy = 0
63. The solubility of KCl is relatively more in:
 (A) C₆H₆ (Dielectric constant = 0)
 (B) (CH₃)₂CO (Dielectric constant = 2)
 (C) CH₃OH (Dielectric constant = 32)
 (D) CCl₄ (Dielectric constant = 0)
64. Lattice energy of an ionic compound depends upon:
 (A) Charge on the ions only
 (B) Size of the ions only
 (C) Packing of the ions only
 (D) Charge and size of the ions
65. Which ion has a higher polarizing power?
 (A) Mg²⁺
 (B) Al³⁺
 (C) Ca²⁺
 (D) Na⁺
- Bond Parameter: Bond Length, Bond Angle, Bond Enthalpy**
66. Most energetic species among the following is
 (A) H₂ (B) Ne
 (C) F (D) F₂
67. The species, having bond angles of 120° is
 (A) ClF₃ (B) NCl₃
 (C) BCl₃ (D) PH₃

68. Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false?
- (A) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is smaller than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
- (B) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 is larger than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
- (C) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 , the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 , and the $\text{H}-\text{O}-\text{H}$ bond angle in H_2O are all greater than 90° .
- (D) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is larger than the $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 .
69. Which of the following has higher bond angle?
- (A) H_2O
 (B) CH_4
 (C) Graphite
 (D) SO_2
70. In BF_3 , the $\text{B}-\text{F}$ bond length is $1.30\text{\AA}$, when BF_3 is allowed to be treated with Me_3N , it forms an adduct, $\text{Me}_3\text{N} \rightarrow \text{BF}_3$, the bond length of $\text{B}-\text{F}$ in the adduct is
- (A) Greater than $1.30\text{\AA}$
 (B) Smaller than $1.30\text{\AA}$
 (C) Equal to $1.30\text{\AA}$
 (D) None of these.
71. The molecule having smallest bond angle is
- (A) AsCl_3 (B) SbCl_3
 (C) PCl_3 (D) NCl_3
72. Decreasing order of bond angle is
- (A) $\text{BeCl}_2 > \text{NO}_2 > \text{SO}_2$
 (B) $\text{BeCl}_2 > \text{SO}_2 > \text{NO}_2$
 (C) $\text{SO}_2 > \text{BeCl}_2 > \text{NO}_2$
 (D) $\text{SO}_2 > \text{NO}_2 > \text{BeCl}_2$
73. The correct arrangement of the species in the decreasing order of the bond length between carbon and oxygen in them is
- (A) $\text{CO}, \text{CO}_2, \text{HCO}_2^-, \text{CO}_3^{2-}$
 (B) $\text{CO}_2, \text{HCO}_2^-, \text{CO}, \text{CO}_3^{2-}$
 (C) $\text{CO}_3^{2-}, \text{HCO}_2^-, \text{CO}_2, \text{CO}$
 (D) $\text{CO}, \text{CO}_3^{2-}, \text{CO}_2, \text{HCO}_2^-$
74. The correct order towards bond angle is
- (A) Bond angle does not depend on hybridization
 (B) $\text{sp} < \text{sp}^2 < \text{sp}^3$
 (C) $\text{sp}^2 < \text{sp} < \text{sp}^3$
 (D) $\text{sp}^3 < \text{sp}^2 < \text{sp}$
75. Which of the following has been arranged in increasing order of % p-character?
- (A) $\text{sp} < \text{sp}^2 < \text{sp}^3$ (B) $\text{sp}^3 < \text{sp}^2 < \text{sp}$
 (C) $\text{sp}^2 < \text{sp}^3 < \text{sp}$ (D) $\text{sp}^2 < \text{sp} < \text{sp}^3$
76. Which of the following compounds have bond angle as nearly 90° ?
- (A) CH_4 (B) CO_2
 (C) H_2O (D) SF_6
77. The bond angle in ethylene is
- (A) 180° (B) 120°
 (C) 109° (D) 90°
78. In which molecule are all atoms coplanar
- (A) CH_4 (B) BF_3
 (C) PF_3 (D) NH_3
79. Which has the least bond angle
- (A) NH_3 (B) BeF_2
 (C) H_2O (D) CH_4
80. In compound X, all the bond angles are exactly $109^\circ 28'$ is
- (A) Chloromethane
 (B) Iodoform
 (C) Carbon tetrachloride
 (D) Chloroform

81. Which of the following molecules has one lone pair of electrons on the central atom
 (A) H_2O (B) NH_3
 (C) CH_4 (D) PCl_5
82. Select the correct order of $\text{H}-\text{M}-\text{H}$ bond angle
 (A) $\text{PH}_3 > \text{PH}_4^+$ (B) $\text{P}_2\text{H}_4 > \text{PH}_4^+$
 (C) $\text{PH}_3 > \text{NH}_4^+$ (D) $\text{PH}_4^+ > \text{NH}_3$
83. Compare bond angle x and y in the following molecules
- 

(A) $x > y$
(C) $x = y$



(B) $y > x$
(D) None of these
84. Find the number of 90° angles in $\text{XeF}_5^\ominus$ molecules
 (A) 1 (B) 2
 (C) 0 (D) 3
85. Find out the relation between (adjacent angle) $\angle\text{FCIF}$ and $\angle\text{FBrF}$ bond angle in ClF_3 and BrF_3 molecule respectively.
 (A) $\angle\text{FCIF} > \angle\text{FBrF}$
 (B) $\angle\text{FCIF} = \angle\text{FBrF}$
 (C) $\angle\text{FCIF} < \angle\text{FBrF}$
 (D) Can't predicted
86. Which one correct for bond angle
 (A) $\text{PF}_3 < \text{PCl}_3$ (B) $\text{PF}_3 < \text{PH}_3$
 (C) $\text{BF}_3 < \text{BCl}_3$ (D) All of these
87. The correct order of $d_{\text{C-H}}$ in the following option is
 (A) $\text{CHF}_3 = \text{CH}_2\text{F}_2 = \text{CH}_3\text{F}$
 (B) $\text{CHF}_3 > \text{CH}_2\text{F}_2 > \text{CH}_3\text{F}$
 (C) $\text{CH}_2\text{F}_2 > \text{CH}_3\text{F} > \text{CHF}_3$
 (D) $\text{CH}_3\text{F} > \text{CH}_2\text{F}_2 > \text{CHF}_3$
88. O_2F_2 is an unstable yellow orange solid and H_2O_2 is a colourless liquid, both have $\text{O}-\text{O}$ bond. $\text{O}-\text{O}$ bond length in H_2O_2 and O_2F_2 is respectively
 (A) $1.22\text{\AA}, 1.48\text{\AA}$ (B) $1.48\text{\AA}, 1.22\text{\AA}$
 (C) $1.22\text{\AA}, 1.22\text{\AA}$ (D) $1.48\text{\AA}, 1.48\text{\AA}$
89. Which of the following statement is incorrect regarding the structure of XeO_2F_2 molecule
 (A) $\text{Xe}=\text{O}$ bonds are present in axial position
 (B) All $\text{Xe}-\text{F}$ bond lengths are identical
 (C) FXeF angles are 90°
 (D) Shape of the molecule is octahedral
90. Comment on the $\text{C}-\text{C}$ bond length for C_2H_6 and C_2F_6 compounds
 (A) $d_{\text{C-C}}(\text{C}_2\text{H}_6) > d_{\text{C-C}}(\text{C}_2\text{F}_6)$
 (B) $d_{\text{C-C}}(\text{C}_2\text{F}_6) > d_{\text{C-C}}(\text{C}_2\text{H}_6)$
 (C) $d_{\text{C-C}}(\text{C}_2\text{F}_6) > d_{\text{C-C}}(\text{C}_2\text{H}_6)$
 (D) Can't be predicted
91. The correct order of ' $\text{S}-\text{O}$ ' bond length is:
 (A) $\text{SO}_3^{2-} > \text{SO}_4^{2-} > \text{SO}_3 > \text{SO}_2$
 (B) $\text{SO}_3^{2-} > \text{SO}_4^{2-} > \text{SO}_2 > \text{SO}_3$
 (C) $\text{SO}_4^{2-} > \text{SO}_3^{2-} > \text{SO}_2 > \text{SO}_3$
 (D) $\text{SO}_4^{2-} > \text{SO}_3^{2-} > \text{SO}_3 > \text{SO}_2$
92. $\angle\text{FAsF}$ bond angle in AsF_3Cl_2 molecule is:
 (A) 90° and 180° (B) 120°
 (C) 90° (D) 180°
93. The correct order of bond angles is:
 (A) $\text{NO}_2^- > \text{NO}_2^+ > \text{NO}_2$
 (B) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
 (C) $\text{NO}_2 > \text{NO}_2^+ > \text{NO}_2^-$
 (D) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$
94. The correct order of $\text{H}-\text{M}-\text{H}$ bonds angle is
 (A) $\text{NH}_3 < \text{PH}_3 < \text{SbH}_3 < \text{BiH}_3$
 (B) $\text{AsH}_3 < \text{SbH}_3 < \text{PH}_3 < \text{NH}_3$
 (C) $\text{NH}_3 < \text{PH}_3 < \text{BiH}_3 < \text{SbH}_3$
 (D) $\text{BiH}_3 < \text{SbH}_3 < \text{AsH}_3 < \text{PH}_3$

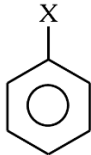
95. The compound MX_4 is tetrahedral. The number of $\angle XMX$ angles in the compound is:
 (A) three (B) four
 (C) five (D) six
96. Which of the following set contains species having same angle around the central atom?
 (A) SF_4, CH_4, NH_3 (B) NF_3, BCl_3, NH_3
 (C) $BF_3, NF_3, AlCl_3$ (D) BF_3, BCl_3, BBr_3
97. In which of the following molecules/ions all the bonds not equal?
 (A) SF_4 (B) SiF_4
 (C) XeF_4 (D) BF_4^-
98. The correct order of bond angles (smallest first) in H_2S, NH_3, BF_3 and SiH_4 is
 (A) $H_2S < SiH_4 < NH_3 < BF_3$
 (B) $H_2S < NH_3 < BF_3 < SiH_4$
 (C) $H_2S < NH_3 < SiH_4 < BF_3$
 (D) $NH_3 < H_2S < SiH_4 < BF_3$

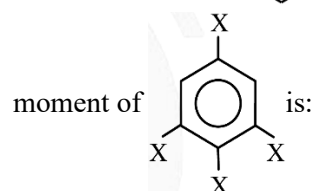
Resonating Structures, Dipole Moment

99. Which of the following molecules has the maximum dipole moment?
 (A) CO_2 (B) CH_4
 (C) NH_3 (D) NF_3
100. Pick up the incorrect statement.
 (A) Dipole moment of ammonia is due to orbital dipole and resultant dipole in the same direction.
 (B) In BF_3 bond dipoles are higher but dipole moment is zero.
 (C) Dipole moment is a vector quantity.
 (D) O_2, H_2 show bond dipole due to polarisation.
101. Which of the following is the correct order of dipole moment?
 (A) $NH_3 < BF_3 < NF_3 < H_2O$
 (B) $BF_3 < NF_3 < NH_3 < H_2O$

- (C) $BF_3 < NH_3 < NF_3 < H_2O$
 (D) $H_2O < NF_3 < NH_3 < BF_3$

- 102.. Among the following molecules, which has the zerodipole moment?
 (A) BF_3 (B) H_2O
 (C) NF_3 (D) ClO_2
103. Which one of the following has the lowest dipole moment?
 (A) CH_3F (B) CH_3Cl
 (C) CH_3I (D) CH_3Br
104. The dipole moment is minimum in
 (A) NH_3 (B) NF_3
 (C) SO_2 (D) BF_3
105. Identify the molecule that has zero dipole moment.
 (A) CH_3Cl
 (B) $CHCl_3$
 (C) CH_2Cl_2
 (D) CCl_4

106. Dipole moment of  in 1.5 D. The dipole



- (A) 1.5 D (B) 2.25 D
 (C) 1 D (D) 3 D

107. The bond length of HCl molecule is $1.275 \text{ \AA}$ and its dipole moment is 1.03 D. The ionic character of the molecule (in percent) (charge of the electron = $4.8 \times 10^{-10} \text{ esu}$) is
 (A) 100 (B) 67.3
 (C) 33.66 (D) 16.83

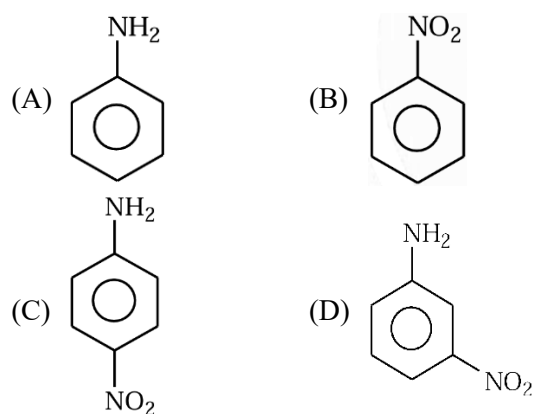
108. Consider the following halogen containing compounds

- (a) CHCl_3 (b) CCl_4
 (c) CH_2Cl_2 (d) 

The compounds with a net zero dipole moment are

- (A) b and d only (B) c only
 (C) c and d only (D) a and d only
109. If $\text{H}-\text{X}$ bond length is $2.00\text{\AA}$ and $\text{H}-\text{X}$ bond has dipole moment $5.12 \times 10^{-30} \text{ C-m}$, the percentage of ionic character in the molecule will be
- (A) 10% (B) 16%
 (C) 18% (D) 20%
110. The dipole moment of HBr is $1.6 \times 10^{-30} \text{ C.m}$ and inter - atomic spacing is $1\text{\AA}$. The % ionic character of HBr is
- (A) 7 (B) 10
 (C) 15 (D) 27
111. If the bond length and dipole moment of a diatomic molecule are $1.25\text{\AA}$ and 1.0 D respectively, what is the per cent ionic character of the bond?
- (A) 10.66 (B) 12.33
 (C) 16.66 (D) 19.33

112. Which of the following will have large dipole moment?



113. Zero dipole moment is possessed by

- (A) PCl_3 (B) BF_3
 (C) ClF_3 (D) NH_3

114. Which one of the following is a non-polar molecule?

- (A) CCl_4 (B) CHCl_3
 (C) CH_2Cl_2 (D) CH_3Cl

115. Identify the non-polar molecule in the set of compounds given $\text{HCl}, \text{HF}, \text{H}_2, \text{HBr}$

- (A) H_2 (B) HCl
 (C) HF, HBr (D) HBr

116. The bond angle and dipole moment of water respectively, are

- (A) $109.5^\circ, 1.84\text{D}$ (B) $107.5^\circ, 1.56\text{D}$
 (C) $104.5^\circ, 1.84\text{D}$ (D) $102.5^\circ, 1.56\text{D}$

117. In a polar molecule, the ionic charge is $4.8 \times 10^{-10} \text{ e.s.u.}$ If the inter ionic distance is $1\text{\AA}$ unit, then the dipole moment is

- (A) 41.8 debye (B) 4.18 debye
 (C) 4.8 debye (D) 0.48 debye

118. The electronegativity of A and B are 1.20 and 4.0 respectively. Therefore, ionic character in A - B bond will be

- (A) 50% (B) 43%
 (C) 53.3% (D) 72.23%

119. If the dipole moment of toluene and nitrobenzene are 0.43 D and 3.93 D , then what is the expected dipole moment of p-nitro toluene?

- (A) 3.50 D (B) 2.18 D
 (C) 4.36 D (D) 5.30 D

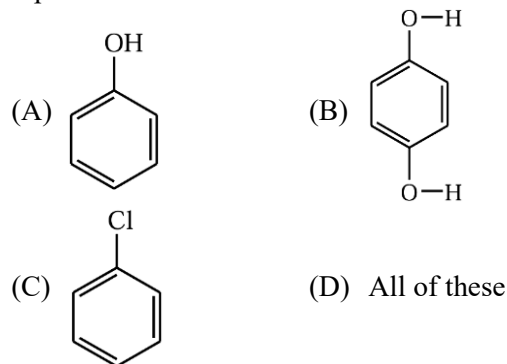
120. Which of the following has dipole moment?

- (A) CO_2
 (B) p-dichlorobenzene
 (C) NH_3
 (D) CH_4

121. The only molecule having dipole moment is

- (A) CCl_4 (B) NH_3
(C) CF_4 (D) BF_3

122. Select the structures which have a permanent dipole moment?



123. The order of polarity of NH_3 , NF_3 and BF_3 is

- (A) $\text{NF}_3 < \text{NH}_3 < \text{BF}_3$
(B) $\text{BF}_3 < \text{NF}_3 < \text{NH}_3$
(C) $\text{BF}_3 < \text{NH}_3 < \text{NF}_3$
(D) $\text{NF}_3 < \text{BF}_3 < \text{NH}_3$

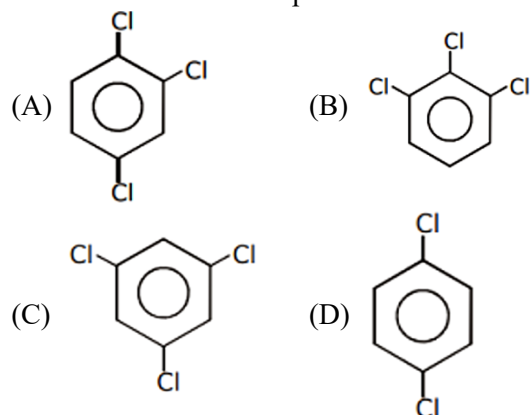
124. The decreasing order of dipole moment of molecules is

- (A) $\text{NF}_3 > \text{NH}_3 > \text{H}_2\text{O}$
(B) $\text{NH}_3 > \text{NF}_3 > \text{H}_2\text{O}$
(C) $\text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3$
(D) $\text{H}_2\text{O} > \text{NF}_3 > \text{NH}_3$

125. Which of the following molecules will have polar bonds but zero dipole moment?

- (A) O_2 (B) CHCl_3
(C) CF_4 (D) None of these

126. Which has maximum dipole moment?



127. Correct set of species with zero dipole moment is:

- (i) CO_2 (ii) COCl_2
(iii) CH_2Cl_2 (iv) BCl_3
(A) (i) and (iv) (B) (ii) and (iv)
(C) (iii) and (iv) (D) (i), (iii) and (iv)

128. H_2O has a net dipole moment while BeF_2 has zero dipole moment because:

- (A) F is more electronegativity than oxygen
(B) Be is more electronegativity than oxygen
(C) H_2O molecule is linear and BeF_2 is bent
(D) BeF_2 molecule is linear and H_2O is bent

129. The dipole moment of HCl is 1.03 D, if $\text{H}-\text{Cl}$ bond distance is $1.26\text{\AA}$, what is the percentage of ionic character in the $\text{H}-\text{Cl}$ bond?

- (A) 60% (B) 39%
(C) 29% (D) 17%

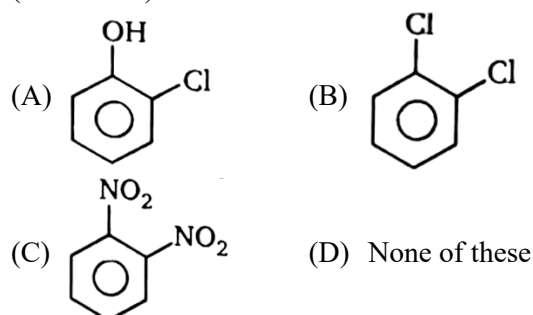
130. The correct order of dipole moment

- (A) $\text{CH}_4 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$
(B) $\text{NF}_3 < \text{CH}_4 < \text{NH}_3 < \text{H}_2\text{O}$
(C) $\text{NH}_3 < \text{NF}_3 < \text{CH}_4 < \text{H}_2\text{O}$
(D) $\text{H}_2\text{O} < \text{NH}_3 < \text{NF}_3 < \text{CH}_4$

131. Which of the following pair of molecules will have permanent dipole moment?

- (A) NO_2 and CO_2 (B) NO_2 and O_3
(C) SiF_4 and CO_2 (D) SiF_4 and NO_2

132. In which of the following molecule μ (observed) is found to be greater than μ (theoretical):



133. Arrange the following compounds in increasing order of dipole moments, toluene (I), odichlorobenzene (II), m-dichlorobenzene (III) and pdichlorobenzene (IV):
 (A) $IV < I < II < III$
 (B) $I < IV < II < III$
 (C) $IV < I < III < II$
 (D) $IV < II < I < III$
134. Which one of the following statements is true?
 (A) The dipole moment of NF_3 is zero.
 (B) The dipole moment of NF_3 is less than NH_3 .
 (C) The dipole moment of NF_3 is more than NH_3 .
 (D) The dipole moment of NH_3 is zero.
135. The dipole moment is the highest for
 (A) trans-2-Butene
 (B) 1, 3-Dimethylbenzene
 (C) Acetophenone
 (D) Ethanol
136. If a molecule MX_3 has zero dipole moment the sigma bonding orbitals used by M (atomic number < 21) are
 (A) Pure p
 (B) sp hybrid
 (C) sp^2 hybrid
 (D) sp^3 hybrid
137. Which of the following arrangement of molecules is correct on the basis of their dipole moments?
 (A) $BF_3 > NF_3 > NH_3$
 (B) $NF_3 > BF_3 > NH_3$
 (C) $NH_3 = NF_3 > BF_3$
 (D) $NH_3 > NF_3 > BF_3$
138. Resonating structures have different
 (A) atomic arrangements
 (B) electronic arrangements
 (C) function groups
 (D) alkyl groups
139. Which among the following resonance structures of N_3^- satisfies the octet rule but is ruled out as a resonance structure?
 (I) $[: \ddot{N} = N = \ddot{N} :]^-$ (II) $[: \ddot{N} - N \equiv N :]^-$
 (III) $[: N \equiv N - \ddot{N} :]^-$ (IV) $\left[\begin{array}{c} :N = N: \\ \diagdown \quad \diagup \\ :N: \end{array} \right]^-$
 (A) IV only
 (B) I and IV only
 (C) I only
 (D) II and III only
140. Resonance is due to
 (A) Delocalization of sigma electrons
 (B) Delocalization of pi electrons
 (C) Migration of H atoms
 (D) Migration of protons
141. For which value of 'x', the dipole moment of the molecule PCl_xF_{5-x} is zero
 (A) 4
 (B) 2
 (C) 3
 (D) 1
142. Of the following molecules, the one, which has permanent dipole moment, is
 (A) SiF_4
 (B) BF_3
 (C) PF_3
 (D) PF_5
143. Which of the following has the least dipole moment?
 (A) NF_3
 (B) SO_3
 (C) XeO_3
 (D) NH_3
144. The dipole moment of the given molecules are such that
 (A) $BF_3 > NF_3 > NH_3$
 (B) $NF_3 > BF_3 > NH_3$
 (C) $NH_3 > NF_3 > BF_3$
 (D) $NH_3 > BF_3 > NF_3$
145. Carbon tetrachloride has no dipole moment because of
 (A) its planar structure
 (B) its regular tetrahedral structure
 (C) similar electron affinities of carbon and chlorine
 (D) none of these

- 146.** The molecule which has zero dipole moment is
 (A) SO_2 (B) PCl_3
 (C) XeF_4 (D) CHCl_3
- 147.** Molecule AB has a bond length of $1.617\text{\AA}$ and a dipole moment of 0.38 D . The fractional charge on each atom (absolute magnitude) is ($e_0 = 4.802 \times 10^{-10}\text{ esu}$)
 (A) 0 (B) 0.05
 (C) 0.5 (D) 1.0
- 148.** Bond-length of HCl is; $1.275\text{\AA}$ ($e = 4.8 \times 10^{-10}\text{ esu}$) If $\mu = 1.02\text{ D}$ then HCl is
 (A) 100% ionic (B) 83% covalent
 (C) 50% covalent (D) 40% ionic
- 149.** Calculate the % ionic character in LiF from following data:
 $\mu_{\text{LiF}} = 6.32\text{ D}$
 $d_{\text{LiF}} = 0.156\text{ nm}$
 (A) 84.32% (B) 20.34%
 (C) 15.68% (D) 100%
- 150.** What is not true about resonance?
 (A) The resonating structures are hypothetical
 (B) The unpaired electrons in various resonating structures are same
 (C) Hybrid structure is most energetic
 (D) Hybrid structure is least energetic
- 151.** Which of the following conditions is not correct for resonating structures?
 (A) The contributing structures must have the same number of unpaired electrons
 (B) The contributing structures should have almost similar energies
 (C) The contributing structures should be so written that unlike charges reside on atoms that are far apart
 (D) The positive charge should be present on the electropositive element and the negative charge on the electronegative element
- 152.** Two hybrid orbitals have a bond angle of 120° . The percentage of s-character in the hybrid orbital is nearly:
 (A) 25% (B) 33%
 (C) 50% (D) 66%
- 153.** Resonance is not shown by
 (A) C_6H_6 (B) NO_2
 (C) CO_3^{2-} (D) H_2O
- 154.** BeF_2 has zero dipole moment whereas H_2O has dipole moment because:
 (A) Water is linear
 (B) H_2O is bent
 (C) F is more electronegative than O
 (D) Hydrogen bonding is present in H_2O
- 155.** Which of the following molecule have zero dipole moment:
 (A) BF_3 (B) CH_2Cl_2
 (C) NF_3 (D) SO_2
- 156.** Which of the following species are polar:
 (a) C_6H_6 (b) XeF_2
 (c) SO_2 (d) SF_4
 (e) SF_6
 Correct answer is:
 (A) (b) & (d) (B) (a), (b) & (e)
 (C) (a) & (e) (D) (c) & (d)
- 157.** Dipole moment is shown by:
 (A) 1, 4-Dichlorobenzene
 (B) Cis 1, 2-dichloroethene
 (C) Trans -1, 2-dichloro ethene
 (D) Benzene
- 158.** What conclusion can be draw from the fact that BF_3 has no dipole moment but PF_3 does:
 (A) BF_3 is not symmetrical but PF_3 is symmetrical
 (B) BF_3 molecule must be linear
 (C) Atomic radius of P is larger than that of B
 (D) BF_3 molecule must be planar triangular

159. Species having zero dipole moment:

- (A) XeF_4 (B) SO_2
(C) SF_4 (D) CH_2Cl_2

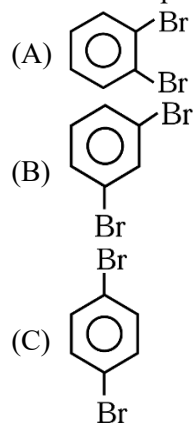
160. Among NO_3^- , CO_3^{2-} , ClO_3^- , SO_3^{2-} and BO_3^{3-} the nonplanar species are -

- (A) CO_3^{2-} , SO_3^{2-} , BO_3^{3-}
(B) ClO_3^- , SO_3^{2-}
(C) NO_3^- , CO_3^{2-} , BO_3^{3-}
(D) NO_3^- , SO_3^{2-} , BO_3^{3-}

161. The molecule BF_3 and NF_3 both are covalent compounds. But BF_3 is non-polar and NF_3 is polar. The reason is that-

- (A) boron is a metal and nitrogen is a gas in uncombined state
(B) BF_3 bonds have no dipole moment whereas NF_3 bonds have dipole moment
(C) atomic size of boron is smaller than that of nitrogen
(D) BF_3 is symmetrical molecule where as NF_3 is unsymmetrical

162. Which of the following molecules shows maximum dipole moment -



(D) All have the same

VSEPR Theory

163. Which of the following molecules has more than one lone pair?

- (A) SO_2 (B) XeF_2
(C) SiF_4 (D) CH_4

164. In the structure of ClF_3 , the number of lone pairs of electrons on central atom 'Cl' is:

- (A) one (B) two
(C) four (D) three

165. The central atom of which one of the following molecules, has 3 bond pairs and 2 lone pairs of electrons in its valence shell?

- (A) Boron trifluoride
(B) Phosphorous trichloride
(C) Chlorine trifluoride
(D) Xenon trioxide

166. Number of b.p. and l.p. in NO_3^- is:

- (A) 3 b.p. +1 l.p. (B) 4 b.p. +0 l.p.
(C) 2 b.p. +2 l.p. (D) 1 b.p. +1 l.p.

167. SF_4 can be classified as

- (A) AB_4 type (B) AB_4E type
(C) AB_4E_2 type (D) none of these

168. Pentagonal bipyramidal structure contains bond angles approximately

- (A) 120° , 90° , 180° (B) 120° , 72° , 180°
(C) 72° , 90° , 120° (D) 72° , 90° , 180°

169. In which of the following molecules, the central atom has two lone pairs of electrons?

- (A) SF_4 (B) BrF_5
(C) SO_2 (D) XeF_4

170. Among the following molecules: SO_2 , SF_4 , ClF_3 , BrF_5 and XeF_4 which of the following shapes does not describe any of the molecules mentioned?

- (A) Bent
(B) Trigonal bipyramidal
(C) See-saw
(D) T-shape

171. In which of the following molecule/ion, all the bonds are not equal?

- (A) XeF_4 (B) BF_4^-
(C) C_2H_4 (D) SiF_4

172. Incorrect matching amongst the following is:

- (i) Linear - $\text{H}_2\text{O}, \text{SO}_2$
 (ii) V-shaped-: $\text{CH}_2, \text{SnCl}_2$
 (iii) See Saw - $\text{SF}_4, \text{TeCl}_4$
 (iv) T-shaped - $\text{ICl}_3, \text{ClF}_3$
 (A) only (i) (B) both (ii) and (iii)
 (C) only (iii) (D) both (iii) and (iv)

173. According to VSEPR theory molecule having Tshape is:

- (A) ClF_3 (B) NH_3
 (C) BF_3 (D) H_2O

174. Which of the following has a regular geometry?

- (A) CHCl_3 (B) PCl_3
 (C) XeF_6 (D) SF_4

175. **Assertion:** ClF_3 has T-shape structure.

Reason: It has two lone pairs arranged at 180° angle.

- (A) If both assertion and reason are true and reason is the correct explanation of assertion.
 (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (C) If assertion is true but reason is false.
 (D) If both assertion and reason are false.

176. Which of the following molecules is linear in shape?

- (A) XeF_6 (B) XeO_3
 (C) XeF_4 (D) XeF_2

177. The shape of XeF_2 is:

- (A) Linear (B) Bent
 (C) T-shaped (D) None of these

178. Which of the following is planar?

- (A) XeO_4 (B) XeO_2F_2
 (C) XeO_3F_2 (D) XeF_4

179. The shape of ICl_5 is:

- (A) triangular bipyramidal
 (B) octahedral
 (C) square pyramidal
 (D) pentagonal planar

180. In $\text{XeF}_2, \text{XeF}_4$ and XeF_6 the number of lone pair of electrons on Xe is respectively.

(Atomic numbers: F = 9, Xe = 54)

- (A) 2, 3, 1 (B) 1, 2, 3
 (C) 4, 1, 2 (D) 3, 2, 1

181. The non-linear structure is assumed by

- (A) SnCl_2 (B) NCO^-
 (C) NO_2^+ (D) CS_2

182. Molecular shapes of SF_4, CF_4 and XeF_4 are:

- (A) same with 2, 0 and 1 lone pairs of electrons respectively.
 (B) same with 1, 1 and 1 lone pairs of electrons respectively.
 (C) different with 1, 0 and 2 lone pairs of electrons respectively.
 (D) different with 1, 1 and 1 lone pairs of electrons respectively.

183. Match List I with List II & select the correct answer:

List I		List II	
A.	ICl_2^-	1.	Linear
B.	BrF_2^+	2.	Pyramidal
C.	ClF_4^-	3.	Tetrahedral
D.	AlCl_4^-	4.	Square planar
		5.	Angular

- (A) (A-1), (B-2), (C-4), (D-5)
 (B) (A-4), (B-5), (C-2), (D-3)
 (C) (A-1), (B-5), (C-4), (D-3)
 (D) (A-5), (B-1), (C-3), (D-4)

184. The shape of XeO_3 is:

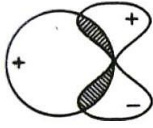
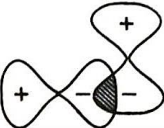
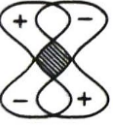
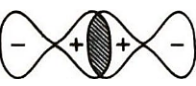
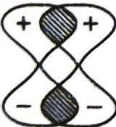
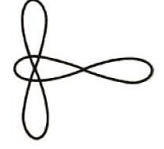
- (A) T-shaped (B) Tetrahedral
 (C) Triangular planar (D) Pyramidal

- 185.** The molecule which has T-shaped structure is:
 (A) BCl_3 (B) NCl_3
 (C) ICl_3 (D) PCl_3
- 186.** The shape of the ICl_3 molecule is:
 (A) Tetrahedral (B) trigonal
 (C) pyramidal (D) T-shaped
- 187.** The geometry around the central atom in ClF_4^+ is
 (A) square planar
 (B) square pyramidal
 (C) octahedral
 (D) trigonal bipyramidal
- 188.** According to VSEPR model, the shape of $[\text{XeOF}_5]^-$ is
 (A) octahedral
 (B) trigonal bipyramidal
 (C) square pyramidal
 (D) pentagonal monopyrarnidal
- 189.** Among SF_4 , BF_4^- , XeF_4 and ICl_4^- the number of species having two lone pair of electrons on the central atom according to VSEPR theory is
 (A) 2 (B) 3
 (C) 4 (D) 0
- 190.** The total number of lone pairs of electrons in I_3^- is
 (A) Zero (B) Three
 (C) Six (D) Nine
- 191.** According to VSEPR theory, the geometry (with lone pair) around the central iodine in I_3^+ and I_3^- ions respectively are
 (A) tetrahedral and tetrahedral
 (B) trigonal bipyramidal and trigonal bipyramidal
 (C) tetrahedral and trigonal bipyramidal
 (D) tetrahedral and octahedral
- 192.** The structure of XeF_2 and XeO_2F_2 respectively are:
 (A) bent, tetrahedral
 (B) linear, square planar
 (C) linear, see-saw
 (D) bent, see-saw
- 193.** The molecule with highest number of lone-pairs and has a linear shape based on VSEPR theory is:
 (A) CO_2 (B) I_3^-
 (C) NO_2^- (D) NO_2^+
- 194.** The number of lone pair(s) of electrons on the central atom in $[\text{BrF}_4]^-$, XeF_6 and $[\text{SbCl}_6]^{3-}$ are, respectively,
 (A) 2,0 and 1 (B) 1,0 and 0
 (C) 2,1 and 1 (D) 2,1 and 0
- 195.** The geometry of the molecule XeO_2F_2 is
 (A) distorted octahedral
 (B) square planar
 (C) trigonal bipyramidal
 (D) tetrahedral
- 196.** Among the following, the isoelectronic and isostructural pair is:
 (A) CO_2 and SO_2 (B) SO_3 and SeO_3
 (C) NO_2^- and TeO_2 (D) SiO_4^{4-} and PO_4^{3-}
- 197.** Match list I (compounds) with list II (structures), and select the correct answer using the codes given below
- | List I | | List II | |
|--------|------------------|---------|-----------------------|
| (a) | XeO_4 | (i) | square planar |
| (b) | BrF_4^- | (ii) | tetrahedral |
| (c) | SeCl_4 | (iii) | Distorted tetrahedral |
- (A) (a - ii) (b - iii) (c - i)
 (B) (a-iii) (b-i) (c-ii)
 (C) (a - ii) (b - i) (c - iii)
 (D) (a - i) (b - ii) (c - iii)

- 198.** Among the following pairs, those in which both species have similar structures are
 (A) N_3^- , XeF_2
 (B) $[\text{ICl}_4]^-$, $[\text{PtCl}_4]^{2-}$
 (C) $[\text{ClF}_2]^+$, $[\text{ClO}_4]^-$
 (D) XeO_3 , SO_3
 (A) (A) and (B) only
 (B) (A) and (C) only
 (C) (A), (B) and (C) only
 (D) (B), (C) and (D) only
- 199.** The number of lone-pairs are identical in the pairs
 (A) XeF_4 , ClF_3
 (B) XeO_4 , ICl_4^-
 (C) XeO_2 , F_2 , ICl_4^-
 (D) XeO_4 , ClF_3
- 200.** The correct non-linear and iso-structural pair is
 (A) SCl_2 , I_3^- (B) SCl_2 , I_3^+
 (C) SCl_2 , ClF_2^- (D) I_3^+ , ClF_2^-
- 201.** According to VSEPR theory, the molecule/ion having ideal tetrahedral shape is
 (A) SF_4
 (B) SO_4^{2-}
 (C) S_2Cl_2
 (D) SO_2Cl_2
- 202.** According to VSEPR theory, the most probable shape of the molecule having 4 electron pairs in the outer shell of the central atom is
 (A) Linear
 (B) Tetrahedral
 (C) Hexahedral
 (D) Octahedral
- 203.** In which of the following species maximum atom can lie in same plane?
 (A) XeF_2O_2 (B) PCl_5
 (C) AsH_4^+ (D) XeF_4
- 204.** Correct statement regarding molecules SF_4 , CF_4 and XeF_4 are:
 (A) 2, 0 and 1 lone pairs of central atom respectively
 (B) 1, 0 and 1 lone pairs of central atom respectively
 (C) 0, 0 and 2 lone pairs of central atom respectively
 (D) 1, 0 and 2 lone pairs of central atom respectively
- 205.** The geometrical arrangement of orbitals and shape of I_3^- are respectively:
 (A) trigonal bipyramidal geometry, linear shape
 (B) hexagonal geometry, T-shape
 (C) trigonal planar geometry, triangular shape
 (D) tetrahedral geometry, pyramidal shape
- 206.** The structure of the noble gas compound XeF_4 is:
 (A) square planar
 (B) distorted tetrahedral
 (C) tetrahedral
 (D) octahedral
- 207.** The molecule exhibiting maximum number of nonbonding electron pairs (l.p.) around the central atom is:
 (A) XeOF_4 (B) XeO_2F_2
 (C) XeF_3^- (D) XeO_3
- 208.** The shape of XeF_4 , XeF_5^- and SnCl_2 are:
 (A) octahedral, trigonal bipyramidal and bent
 (B) square pyramidal, pentagonal planar and linear
 (C) square planar, pentagonal planar and angular
 (D) see-saw, T-shaped and linear
- 209.** Which is not correctly matched?
 (A) XeO_3 - Trigonal bipyramidal
 (B) ClF_3 - bent T-shape
 (C) XeOF_4 - Square pyramidal
 (D) XeF_2 - Linear shape

210. What is the shape of the ClF_3 molecule?
- (A) Trigonal planar
(B) Trigonal pyramidal
(C) T-shaped
(D) Tetrahedral
211. According to VSEPR theory, in which species do all the atoms lie in the same plane?
1. CH_3^+ 2. CH_3^-
(A) 1 only (B) 2 only
(C) both 1 and 2 (D) neither 1 nor 2
212. Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false:
- (I) The order of repulsion between different pair of electrons is $\text{lp-lp} > \text{lp-bp} > \text{bp-bp}$
(II) In general, as the number of lone pair of electrons on central atom increases, value of bond angle from normal bond angle also increases.
(III) The number of lone pair on O in H_2O is 2 while on N in NH_3 is 1.
(IV) The structures of xenon fluorides and xenon oxyfluorides could not be explained on the basis of VSEPR theory.
- (A) T T T F (B) T F T F
(C) T F T T (D) T F F F
213. The pair of species having identical shapes for molecules of both species is:
- (A) $\text{XeF}_2, \text{CO}_2$ (B) $\text{BF}_3, \text{PCl}_3$
(C) PF_5, IF_5 (D) CF_4, SF_4
214. Which of the following is not applicable -
- (A) CO_2 Irregular geometry
(B) BF_3 Regular geometry
(C) NH_3 Irregular geometry
(D) SO_2 Irregular geometry
215. The shape of sulphate ion is -
- (A) hexagonal
(B) square planar
(C) trigonal bipyramidal
(D) tetrahedral

Valence Bond Theory (VBT) : Directional properties of bonds, Overlapping of Atomic Orbitals, Types of overlapping and nature of covalent bonds

216. The C-H bond and C-C bond in ethane are formed by which of the following types of overlap?
- (A) $\text{sp}^3 - \text{s}$ and $\text{sp}^3 - \text{sp}^3$
(B) $\text{sp}^2 - \text{s}$ and $\text{sp}^2 - \text{sp}^2$
(C) $\text{sp} - \text{s}$ and $\text{sp} - \text{p}$
(D) $\text{p} - \text{s}$ and $\text{p} - \text{p}$
217. Which one of the following molecules are formed by p-p overlapping?
- (A) Cl_2 (B) HCl
(C) H_2O (D) NH_3
218. Which of the following atomic orbital overlappings are not allowed?
- (P)  (Q) 
(R)  (S) 
(T) 
- (A) All of these (B) (P)(Q)(R)
(C) (P) (R) (T) (D) (Q) only
219. Which p-orbitals overlapping would give the strongest bond?
- (A)  (B) 
(C)  (D) 

220. The bond in the formation of fluorine molecule will be
 (A) Due to s-s overlapping
 (B) Due to s-p overlapping
 (C) Due to p-p overlapping
 (D) Due to hybridization

221. Which type of overlapping results the formation of a π bond
 (A) Axial overlapping of s-s orbitals
 (B) Lateral overlapping of p-p orbitals
 (C) Axial overlapping of p-p orbitals
 (D) Axial overlapping of s-p orbitals

222. In a sigma bond
 (A) Sidewise as well as end to end overlap of orbitals take place
 (B) Sidewise overlap of orbitals takes place
 (C) End to end overlap of orbitals takes place
 (D) None of the above

223. Which is correct option for overlapping strength
 (A) $2s-2s < 2s-2p < 2p-2p$
 (B) $1s-1s < 1s-2p < 1p-2p$
 (C) $2s-2s < 1s-2p < 1s-1s$
 (D) Both (A) and (C)

224. Which overlapping consist of maximum overlapped area between nucleus
 (A) $2s-2s$
 (B) $2p-2p$
 (C) $2s-2p$
 (D) Both (A) and (C)

Hybridization

225. A molecule MX_4 has a square planar shape. The number of non-bonding pair of electrons is
 (A) 2 (B) 1
 (C) 3 (D) 0
226. The percentage of s-character in the hybridized orbitals of B in BF_3 is:
 (A) 25 (B) 50
 (C) 75 (D) 33.3

227. In which of the following pairs both the species have sp^3 hybridization?

- (A) SiF_4, BeH_2 (B) NF_3, H_2O
 (C) NF_3, BF_3 (D) H_2S, BF_3

228. Which of the following statements is correct?

- (A) sp^3 hybrid orbitals have equal s and p character.
 (B) The bond angle decreases with the decrease of s character of hybridized orbital.
 (C) Resonance decreases the stability of a molecule.
 (D) Resonance is due to delocalization of sigma electrons.

229. The percentage s-character of the hybrid orbitals in methane, ethene and ethyne are respectively:

- (A) 25,33,50 (B) 25,50,75
 (C) 50,75,100 (D) 10,20,40

230. The percentage of p-character of the hybrid orbitals in graphite and diamond are respectively:

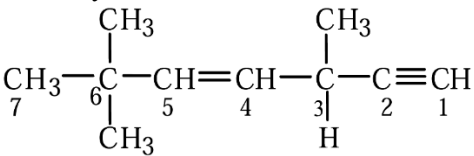
- (A) 33 and 25 (B) 50 and 75
 (C) 67 and 75 (D) 33 and 75

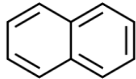
231. Pick out the incorrect statement from the following.

- (A) sp hybrid orbitals are equivalent and are at an angle of 180° with each other.
 (B) sp^2 hybrid orbitals are equivalent and bond angle between any two of them is 120°
 (C) sp^3d^2 hybrid orbitals are equivalent and are oriented towards corners of a regular octahedron.
 (D) sp^3d^3 hybrid orbitals are not equivalent.

232. Which of the following has maximum % of s-character?

- (A) N_2H_2 (B) N_2H_4
 (C) NH_3 (D) NH_2^-

- 233.** The state of hybridization of the central atom and the number of lone pairs over the central atom in POCl_3 are
 (A) $\text{sp}, 0$ (B) $\text{sp}^2, 0$
 (C) $\text{sp}^3, 0$ (D) $\text{dsp}^2, 1$
- 234.** Which of the following is incorrect about cyanogen gas?
 (A) It has bent structure
 (B) It is pseudohalogen
 (C) Its behaviour is similar to halides
 (D) Both carbons are sp hybridized
- 235.** What is the percentage of s -subshell in H_2O molecule?
 (A) 33.33% (B) 50%
 (C) 25% (D) 75%
- 236.** The AsF_5 molecule is trigonal bipyramidal. The hybrid orbitals used by the As atoms for bonding are:
 (A) $\text{d}_{x^2-y^2}, \text{d}_{z^2}, \text{s}, \text{p}_x, \text{p}_y$ (B) $\text{u}_{xy}, \text{s}, \text{p}_x, \text{p}_y, \text{p}_z$
 (C) $\text{d}_{x^2-y^2}, \text{s}, \text{p}_x, \text{p}_y$ (D) $\text{s}, \text{p}_x, \text{p}_y, \text{p}_z, \text{d}_{z^2}$
- 237.** The hybridization at carbons involved in C^*-C^* bond in $\text{HC}\equiv\text{C}^*-\text{CH}^*=\text{CH}_2$ is:
 (A) sp^3-sp^2 (B) sp^3-sp^3
 (C) $\text{sp}-\text{sp}^2$ (D) $\text{sp}-\text{sp}^3$
- 238.** The hybridisation and shape of ClO_3^- is:
 (A) sp^3 and tetrahedral
 (B) sp^3 and triangular pyramidal
 (C) sp^3 and triangular planar
 (D) sp^3d and trigonal bipyramidal
- 239.** The state of hybridization of Xe in XeF_4 is:
 (A) sp^2 (B) sp^3
 (C) sp^3d (D) sp^3d^2
- 240.** The hybridization of S in HSO_3^- and shape is:
 (A) sp^3 and pyramidal
 (B) sp^2 and trigonal planar
 (C) sp^3 and tetrahedral
 (D) sp^3d and trigonal bipyramidal.
- 241.** In piperidine  N -atom has hybridization:
 (A) sp (B) sp^2
 (C) sp^3 (D) dsp^2
- 242.** Which hybridization results non-planar orbitals?
 (A) sp (B) sp^2
 (C) sp^3 (D) dsp^2
- 243.** In the following molecule, the two carbon atoms marked by asterisk (*) possess the following type of hybridized orbitals:
 $\text{H}_3\text{C}-\text{C}^*\equiv\text{C}^*-\text{CH}_3$
 (A) sp^3 -orbital (B) sp^2 -orbital
 (C) sp -orbital (D) s -orbital
- 244.** The hybridization of P in PO_4^{3-} is same as in:
 (A) I in ICl_4^- (B) S in SO_3
 (C) N in NO_3^- (D) S in SO_4^{2-}
- 245.** The state of hybridization of $\text{C}_2, \text{C}_3, \text{C}_5$ and C_6 of the hydrocarbon,

 is in the following sequence:
 (A) $\text{sp}, \text{sp}^2, \text{sp}^3$ and sp^2
 (B) $\text{sp}, \text{sp}^3, \text{sp}^2$ and sp^3
 (C) $\text{sp}^3, \text{sp}^2, \text{sp}^2$ and sp
 (D) $\text{sp}, \text{sp}^2, \text{sp}^2$ and sp^3

246. In which of the following pairs of molecules/ions, the central atom has sp^2 -hybridization?
 (A) NO_2 and NH_3
 (B) BF_3 and NO_2^-
 (C) NH_2^- and H_2O
 (D) BF_3 and NH_2^+
247. sp^2 -hybridization is shown by:
 (A) $BeCl_2$ (B) BF_3
 (C) NH_3 (D) XeF_2
248. Which does not use sp^3 -hybrid orbitals in its bonding?
 (A) BeF_3^- (B) OH_3^+
 (C) NH_4^+ (D) NF_3
249. Carbon atoms in $C_2(CN)_4$ are:
 (A) sp -hybridized
 (B) sp^2 -hybridized
 (C) sp and sp^2 -hybridized
 (D) sp, sp^2 and sp^3 -hybridized
250. Which has sp^2 -hybridization?
 (A) CO_2 (B) SO_2
 (C) N_2O (D) CO
251. The hybridization of S in HSO_3^- and shape is:
 (A) sp^3 and pyramidal
 (B) sp^2 and trigonal planar
 (C) sp^3 and tetrahedral
 (D) sp^3d and trigonal bipyramidal
252. In the molecules H_2O, NH_3 and CH_4 .
 (A) The bond angles are same
 (B) The bond distances are same.
 (C) The hybridizations are same
 (D) The shapes are same
253. Among the following which species has same number of σ and π -bond?
 (A) C_7H_8 (B) $C_2(CN)_4$
 (C) C_2H_4 (D) $HC \equiv CH$
254. Number of π -bonds in naphthalene is:

 (A) 6 (B) 3
 (C) 4 (D) 5
255. Which one of the following is the correct set with respect to molecule, hybridization and shape?
 (A) $BeCl_2, sp^2$, linear
 (B) $BeCl_2, sp^2$, triangular planar
 (C) BCl_3, sp^2 , triangular planar
 (D) BCl_3, sp^3 , tetrahedral
256. The number of sp^2-s sigma bonds in benzene are:
 (A) 3 (B) 6
 (C) 12 (D) None of these
257. The hybridization of central iodine atom in IF_5, I_3^- and I_3^+ are respectively:
 (A) sp^3d^2, sp^3d, sp^3 (B) sp^3d, sp^3d, sp^3
 (C) sp^3d^2, sp^3d^2, sp^3 (D) sp^3d, sp^3d^2, sp^3
258. The species in which the N atom is in a state of sp hybridization is:
 (A) NO_2^+ (B) NO_2^-
 (C) NO_3^- (D) NO_2
259. Which of the following statement is not correct?
 (A) Hybridisation is the mixing of atomic orbitals and their simultaneous combination into molecular orbitals
 (B) sp^2 hybrid orbitals are formed from two p atomic orbitals and one s-atomic orbital.
 (C) dsp^2 hybrid orbitals are all at 90° to one another
 (D) d^2sp^3 hybrid orbitals are directed towards the corners of a regular octahedron

- 260.** Which carbon is more electronegative?
 (A) sp^3 hybridized carbon
 (B) sp -hybridized carbon
 (C) sp^2 hybridized carbon
 (D) Always same irrespective of its hybrid state
- 261.** On hybridization of one s and one p orbital we get
 (A) Two mutually perpendicular orbitals
 (B) Two orbitals at 180°
 (C) Four orbitals directed tetrahedrally
 (D) Three orbitals in a plane
- 262.** Which of the following represents the given mode of hybridization $sp^2-sp^2-sp-sp$ from left to right?
 (A) $H_2C=CH-C\equiv N$
 (B) $HC\equiv C-C\equiv CH$
 (C) $H_2C=C=C=CH_2$
 (D) $HC\equiv CH$
- 263.** Which is correct statement?
 As the s-character of a hybrid orbital decreases
 (I) The bond angle decreases
 (II) The bond strength increases
 (III) The bond length increases
 (IV) Size of orbitals increases
 (A) (I), (III) and (IV)
 (B) (II), (III) and (IV)
 (C) (I) and (II)
 (D) All are correct
- 264.** BF_3 and NF_3 both are covalent compounds but NF_3 is polar whereas BF_3 is non-polar. This is because:
 (A) Nitrogen atom is smaller than boron atom
 (B) $N-F$ bond is more polar than $B-F$ bond
 (C) NF_3 is pyramidal whereas BF_3 is planar triangular
 (D) BF_3 is electron deficient whereas NF_3 is not
- 265.** Position of fluorine in PCl_3F_2
 (A) Both at equatorial position
 (B) Both at axial position
 (C) 1 at equatorial and 1 at axial position
 (D) Cannot be predicted
- 266.** In sp^3d hybridization with 2 lone pairs, the lone pairs are placed at ____ position and shape of molecule is ____ .
 (A) Equatorial, trigonal bipyramidal
 (B) Axial, trigonal bipyramidal
 (C) Equatorial, Bent-T
 (D) Axial, Trigonal planar
- 267.** %S character in equatorial orbital in TBP is:
 (A) 50% (B) 25%
 (C) 33.33% (D) 0%
- 268.** Order of C-H bond length in CH_4 and CH_2F_2 is:
 (A) $CH_4 > CH_2F_2$
 (B) $CH_2F_2 > CH_4$
 (C) $CH_4 = CH_2F_2$
 (D) Cannot say anything
- 269.** Minimum bond angle is present in
 (A) NH_3 (B) CH_4
 (C) H_2O (D) H_2Se
- 270.** Compare bond angle among H_2O, OF_2, OMe_2 and OCl_2
 (A) $H_2O < OF_2 < OMe_2 < OCl_2$
 (B) $OF_2 < H_2O < OCl_2 < OMe_2$
 (C) $OCl_2 < OMe_2 < H_2O < OF_2$
 (D) $OCl_2 < OF_2 < OMe_2 < H_2O$
- 271.** Which of the compound has the longest S-O bond length?
 (A) $O=SF_2$
 (B) $O=SCl_2$
 (C) $O=SBr_2$
 (D) All are equal

272. Relation between % s character in hybrid orbitals and bond length is:
 (A) Equal
 (B) Directly proportional
 (C) Inversely proportional
 (D) No relation
273. Bond angle in PH_3 is:
 (A) Close to 109.28° (B) Same as NH_3
 (C) Close to 90° (D) Close to 120°
274. Lone pairs in H_2Se are present in
 (A) Sp^3 hybrid orbital
 (B) pure p & s orbital
 (C) pure s orbital
 (D) sp^2 hybrid orbital
275. In which of the following compounds B–F bond length is shortest?
 (A) BF_4^-
 (B) $\text{BF}_3 \leftarrow \text{NH}_3$
 (C) BF_3
 (D) $\text{BF}_3 \leftarrow \text{N}(\text{CH}_3)_3$
276. "Hybridisation of central atom does not always change due to back bonding". This statement is valid for which of the following compounds?
 (i) CCl_3^- (ii) CCl_2
 (iii) $(\text{SiH}_3)_2\text{O}$ (iv) $\text{N}(\text{SiH}_3)_3$
 (A) (i), (ii) (B) (i), (iii)
 (C) (ii), (iii) (D) All
277. PCl_5 and PBr_5 exist in sp^3d hybrid state in gaseous phase but in solid state, which of the following statement is true?
 (A) P in PCl_5 exists in sp^3 -hybridization state, while P in PBr_5 exists in sp^3d^2 and sp^3 hybridization states
 (B) P in PCl_5 and PBr_5 exists in sp^3d^2 and sp^3 hybridization state
 (C) P in PCl_5 exists in sp^3d^2 and sp^3 hybridization states, while P in PBr_5 exists in sp^3 -hybridization state
 (D) P in PCl_5 and PBr_5 exists in sp^3 -hybridization state
278. Hybridisation of N_2O_5 in the solid state is:
 (A) $\text{sp}^3, \text{sp}^3\text{d}^2$
 (B) sp, sp^2
 (C) $\text{sp}^3, \text{sp}^3\text{d}$
 (D) sp, sp^3
279. In which of the following molecule back bonding is present,
 (A) BF_3 (B) H_2O
 (C) NH_3 (D) HF
280. How many resonance forms can be written for the nitrate ion, (NO_3^-) ?
 (A) 1 (B) 2
 (C) 3 (D) 4
281. Hybridisation of PBr_5 in the solid state is:
 (A) sp^3 (B) sp^3d
 (C) sp^2 (D) sp
282. The number and type of bonds between two carbon atoms in calcium carbide are
 (A) One sigma, one pi
 (B) One sigma, two pi
 (C) Two sigma, one pi
 (D) Two sigma, two pi
283. Strongest bond is
 (A) $\text{C}-\text{C}$
 (B) $\text{C}=\text{C}$
 (C) $\text{C}\equiv\text{C}$
 (D) All are equally strong
284. π bond is formed
 (A) By overlapping of atomic orbitals on the axis of nuclei
 (B) By mutual sharing of pi electron
 (C) By sidewise overlapping of half-filled p-orbitals
 (D) By overlapping of s-orbitals with p-orbitals

285. Choose the molecules in which hybridisation occurs in the ground state?

- (A) BCl_3 (B) NH_3
(C) PCl_3 (D) BeF_2

The correct answer is

- (A) A,B,D (B) A,B,C
(C) B,C (D) C,D

286. The hybridization in PF_3 is

- (A) sp^3 (B) sp^2
(C) dsp^3 (D) d^2sp^3

287. The hybridisation of carbon in diamond, graphite and acetylene is

- (A) $\text{sp}^3, \text{sp}^2, \text{sp}$ (B) $\text{sp}^3, \text{sp}, \text{sp}^2$
(C) $\text{sp}^2, \text{sp}^3, \text{sp}$ (D) $\text{sp}, \text{sp}^3, \text{sp}^2$

288. Each carbon in carbon suboxide is

- (A) sp^2 -hybridized
(B) sp^3 -hybridized
(C) sp -hybridized
(D) sp^3 -hybridized but linked with one coordinate bond

289. Among the following pairs in which the two species are not isostructural is

- (A) SiF_4 and SF_4 (B) IO_3^- and XeO_3
(C) BH_4^- and NH_4^+ (D) PF_6^- and SF_6

290. Which molecule is not linear

- (A) BeF_2 (B) BeH_2
(C) CO_2 (D) H_2O

291. The bond angle in water molecule is nearly or Directed bonds in water forms an angle of

- (A) 120° (B) 180°
(C) $109^\circ 28'$ (D) $104^\circ 30'$

292. The central atom in a molecule is in sp^2 hybrid state. The shape of molecule will be

- (A) Pyramidal (B) Tetrahedral
(C) Octahedral (D) Trigonal planar

293. Which molecule is linear

- (A) NO_2 (B) ClO_2
(C) CO_2 (D) H_2S

294. Which of the following molecules has trigonal planar geometry

- (A) IF_3 (B) PCl_3
(C) NH_3 (D) BF_3

295. A sp^3 hybridized orbital contains

- (A) $\frac{1}{4}$ s - character (B) $\frac{1}{2}$ s - character
(C) $\frac{2}{3}$ s - character (D) $\frac{3}{4}$ s - character

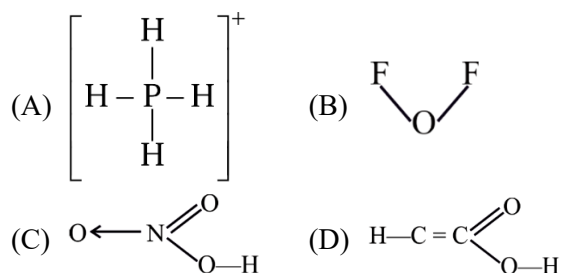
296. Structure of ammonia is

- (A) Trigonal
(B) Tetrahedral
(C) Pyramidal
(D) Trigonal bipyramidal

297. Compound formed by sp^3d hybridization will have structure

- (A) Planar
(B) Pyramidal
(C) Angular
(D) Trigonal bipyramidal

298. Which of the following formula does not correctly represent the bonding capacity of the atom involved



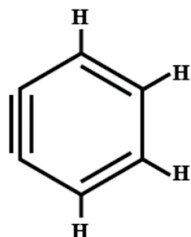
299. The mode of hybridisation of carbon in CO_2 is

- (A) sp (B) sp^2
(C) sp^3 (D) None of these

- 300.** In which of the following the central atom does not use sp^3 hybrid orbitals in its bonding
 (A) BeF_3^- (B) OH_3^+
 (C) NH_2^- (D) NF_3
- 301.** Of the following compounds, the one having a linear structure is
 (A) NH_3 (B) CH_4
 (C) XeF_2 (D) H_2O
- 302.** XeF_6 is
 (A) Octahedral
 (B) Distorted octahedral
 (C) Planar
 (D) Tetrahedral
- 303.** Of the following species the one having a square planar structure is
 (A) NH_4^+ (B) BF_4^-
 (C) XeF_4 (D) SCl_4
- 304.** In which of the following is the angle between the two covalent bonds greatest
 (A) CO_2 (B) CH_4
 (C) NH_3 (D) H_2O
- 305.** Which of the following molecules does not have a linear arrangement of atoms
 (A) H_2S (B) I_3^-
 (C) BeH_2 (D) CO_2
- 306.** What is the percentage of s-character in the orbital occupied by the lone pair of NH_3 . It is given that: $\cos 107^\circ = -.292$
 (A) 22.6% (B) 32.2%
 (C) 25% (D) None of these
- 307.** Maximum number of atom(s) in I_2Cl_6 which is/are in the same plane.
 (A) 4 (B) 8
 (C) 6 (D) 2
- 308.** The maximum number of atoms may lie on one plane of $P(CH_3)_3(CF_3)_2$ is
 (A) 4 (B) 5
 (C) 6 (D) 7
- 309.** The maximum % of S-character in N-H bond is observed in
 (A) NH_3 (B) NH_4^+
 (C) N_2H_4 (D) N_2H_2
- 310.** Consider the following statements and select the correct set of codes using T(True) and F(False) in the given sequence.
 (i) $PI_5(s)$ does not exist due to steric crowding.
 (ii) All possible angle in BF_2Cl are 120° .
 (iii) In $N(SiH_3)_3$ lone pair of N is present in sp^3 hybrid orbital.
 (iv) 'd' orbital used in hybridization of 'P' in $PBr_5(s)$ is d_{z^2}
 (A) TFFT (B) FTTT
 (C) TFFF (D) TFTF
- 311.** Maximum seven atoms are present in a plane. Except in
 (A) H_2CSF_4 (B) $P(CF_3)_2(CH_3)_3$
 (C) $N(SiH_3)_3$ (D) $B(CH_3)_3$
- 312.** The strongest P–O bond is found in the molecule
 (A) F_3PO (B) Cl_3PO
 (C) Br_3PO (D) $(CH_3)_3PO$
- 313.** Select the incorrect statement(s) about N_2F_4 and N_2H_4 .
 (A) In N_2F_4 , d-orbitals are contracted by electronegative fluorine atoms, but d-orbital contraction is not possible by H-atom in N_2H_4
 (B) The N–N bond energy in N_2F_4 is more than N–N bond energy in N_2H_4
 (C) The N–N bond length in N_2F_4 is more than that of N_2H_4
 (D) The N–N bond length in N_2F_4 is less than that of N_2H_4

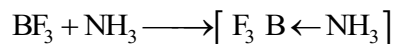
- 314.** The correct statement about the structures of H_2O_2 , O_2F_2 and OF_2 is/are
- (A) H_2O_2 , O_2F_2 , OF_2 are polar compounds
 (B) $d_{\text{O-O}}$ of $\text{H}_2\text{O}_2 > d_{\text{O-O}}$ of O_2F_2
 (C) $d_{\text{O-F}}$ of $\text{OF}_2 > d_{\text{O-F}}$ of O_2F_2
 (D) the strength of O-O bond in O_2F_2 is greater than that of H_2O_2
- 315.** Introducing the Cl -atoms in PF_5 we may get molecule $\text{PCl}_x\text{F}_{5-x}$
 When the value of $x=2$. Comment on the position of F -atoms in the molecule.
 (A) Three F-atoms are the three-equatorial position
 (B) Two F-atoms are at the equatorial position and one F -atom is at the axial position
 (C) Two F -atoms are at axial position and one F atom is at the equatorial position
 (D) All are correct
- 316.** Which is the true statement about $(\text{SiH}_3)_3\text{N}$?
- (A) It is trigonal planar
 (B) It is trigonal pyramidal
 (C) It is stronger lewis base than that of $(\text{CH}_3)_3\text{N}$
 (D) It has a total of 9 sigma bonds
- 317.** For BF_3 molecule which of the following is true?
- (A) B-atom is sp^2 hybridised
 (B) There is a $\text{p}\pi-\text{p}\pi$ back bonding in this molecule
 (C) Observed B-F bond length is found to be less than the expected bond length
 (D) All of these
- 318.** Which of the following statements regarding the structure of SOCl_2 is not correct?
- (A) The sulphur is sp^3 hybridised and it has a tetrahedral shape
 (B) The sulphur is sp^3 hybridised and it has a trigonal pyramid shape
 (C) The oxygen-sulphur bond is $\text{p}\pi-\text{d}\pi$ bond
 (D) It contains one lone pair of electrons in the sp^3 hybrid orbital of sulphur
- 319.** Respective order of strength of back-bonding in boron trihalides is
- (A) $\text{BF}_3 < \text{BCl}_3 < \text{BBR}_3$
 (B) $\text{BF}_3 > \text{BCl}_3 > \text{BBR}_3$
 (C) $\text{BBR}_3 > \text{BCl}_3 > \text{BF}_3$
 (D) $\text{BF}_3 < \text{BCl}_3 < \text{BBR}_3$
- 320.** For BF_3 molecule which of the following will not be true
- (A) It has less bond length than BF_4^-
 (B) It has less bond length than the compound $[\text{NH}_3\text{BF}_3]$
 (C) Its bond strength is increased because of $\text{p}\pi-\text{d}\pi$ back bonding
 (D) It forms BF_4^- when hydrolysed in water
- 321.** Out of (1) SiH_3NCO and (2) CH_3NCO is non-linear
- (A) only 1 (B) only 2
 (C) both 1 and 2 (D) none of these
- 322.** Which of the following option are correct regarding back bonding
- (A) It is an intramolecular lewis acid and lewis base attraction
 (B) It is an intermolecular lewis acid and lewis base attraction
 (C) Both are correct
 (D) It is an intermolecular lewis acid and lewis acid attraction
- 323.** As a result of sp hybridization, we get
- (A) Two mutual perpendicular orbitals
 (B) Two orbitals at 180°
 (C) Four orbitals in tetrahedral directions
 (D) Three orbitals in the same plane

324. How many sp^2 and sp -hybridised carbon atoms are present respectively in the following compound?



- (A) 4, 2 (B) 6, 0
(C) 3, 3 (D) 5, 1
325. $CH_3 - CH_2 - CH = CH_2$ has hybridization:
(A) sp, sp, sp^2, sp^2
(B) sp^2, sp^3, sp^2, sp
(C) sp^3, sp^3, sp^2, sp^2
(D) sp^3, sp^2, sp^2, sp
326. During the complete combustion of methane CH_4 , what change in hybridization does the carbon atom undergo?
(A) sp^3 to sp (B) sp^3 to sp^2
(C) sp^2 to sp (D) sp^2 to sp^3
327. Which bonds are formed by a carbon atom with sp^2 hybridisation?
(A) 4π -bonds
(B) 2π -bonds and 2σ -bonds
(C) 1π -bonds and 3σ -bonds
(D) 4σ -bonds
328. A molecule has trigonal bipyramidal shape then which d orbital participate in hybridization.
(A) d_{xy} (B) $d_{x^2-y^2}$
(C) d_{z^2} (D) d_{zx}
329. PCl_5 exists but NCl_5 does not because
(A) Nitrogen has no vacant 2d-orbitals
(B) NCl_5 is unstable
(C) Nitrogen atom is much smaller than P
(D) Nitrogen is highly inert

330. Correct statement regarding this reaction



- (A) Hybridisation of N is changed
(B) Hybridisation of B is changed
(C) NH_3 act as a lewis base
(D) (B) & (C) both
331. The correct statement for the following reaction is:
 $NH_3 + H^+ \rightarrow NH_4^+$
(A) Hybridisation state is changed
(B) Bond angle increases
(C) NH_3 act as a Lewis acid
(D) Regular geometry is changed
332. In which of the following species is the underlined carbon having sp^3 -hybridisation?
(A) $CH_3 - \underline{C}OOH$ (B) $CH_3 \underline{C}H_2OH$
(C) $CH_3 \underline{C}OCH_3$ (D) $CH_2 = \underline{C}H - CH_3$
333. Which of the following is not true for a hybridisation process -
(A) Electrons take part and not the orbitals
(B) Participating orbitals have almost equal energy
(C) Hybridised orbitals have identical shape
(D) Hybridised orbitals have identical energies
334. The bond between carbon atom (1) and carbon atom (2) in compound,
 $N \equiv \underset{(1)}{C} - \underset{(2)}{CH} = CH_2$ involves the hybrid as
(A) sp and sp^2 (B) sp^2 and sp^3
(C) sp and sp^3 (D) sp and sp
335. The relative overlap of orbitals decreases in the order -
(A) $sp > sp^2 > sp^3 \gg p$
(B) $sp^3 > sp^2 > sp \gg p$
(C) $p > sp^3 > sp^2 > sp$
(D) $p > sp > sp^2 > sp^3$

336. Choose the molecules in which hybridisation occurs in the ground state -

- (a) BCl_3 (b) NH_3
(c) PCl_3 (d) BeF_2

The correct answer is -

- (A) a, b, d (B) a, b, c
(C) b, c (D) c, d

337. What is not true for SiH_4 molecule -

- (A) Tetrahedral geometry
(B) 109° angle
(C) 4σ bond
(D) 4-lone pair of electrons

Molecular Orbital Theory : Nature of the bond, Bond-length, Magnetic nature, Bond Order Calculation, Bond Dissociation Energy

338. The bond length in O_2^+ , O_2 , O_2^- and O_2^{2-} follows the order

- (A) $\text{O}_2^{2-} > \text{O}_2^- > \text{O}_2 > \text{O}_2^+$
(B) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
(C) $\text{O}_2 > \text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+$
(D) $\text{O}_2^- > \text{O}_2^{2-} > \text{O}_2^+ > \text{O}_2$

339. Which of the following of pair species has the bond order equal to $1/2$?

- (A) H_2, H_2^+ (B) $\text{He}_2^+, \text{He}_2$
(C) $\text{He}_2, \text{H}_2^+$ (D) $\text{H}_2^+, \text{He}_2^+$

340. The total number of antibonding electrons in valence shell of NO and CO according to MO theory are respectively.

- (A) 3, 2 (B) 2, 2
(C) 1, 0 (D) 2, 3

341. If a homonuclear diatomic molecule is oriented along the Z-axis, the molecular orbital formed by linear combination of p_x , orbitals of the two atoms is

- (A) σ (B) σ^*
(C) π (D) δ

342. The molecule in which the bond order increases upon addition of an electron is

- (A) O_2 (B) B_2
(C) P_2 (D) N_2

343. The reason for the chemical inertness of gaseous nitrogen at room temperature is best given by its

- (A) high bonding energy only
(B) electronic configuration
(C) HOMO-LUMO gap only
(D) high bond energy and HOMO-LUMO gap

344. The correct statement among the following is

- (A) N_2 has higher bond order than N_2^+ and hence has larger bond length compared to N_2^+
(B) N_2^+ has higher bond order than N_2 and hence has larger bond length compared to N_2
(C) N_2 has higher bond order than N_2^+ and hence has higher dissociation energy compared to N_2^+
(D) N_2 has lower bond order than N_2^+ and hence has lower dissociation energy compared to N_2^+ energy

345. The highest occupied MO in N_2 and O_2^+ respectively are (take x -axis as internuclear axis)

- (A) $\sigma 2p_x, \pi^* 2p_y$
(B) $\pi 2p_y, \pi 2p_z$
(C) $\sigma^* 2p_x, \sigma 2p_x$
(D) $\pi^* 2p_y, \pi^* 2p_z$

346. A molecular orbital of a diatomic molecule changes sign when it is rotated by 180° around the molecular axis. This orbital is

- (A) σ (B) π
(C) δ (D) ϕ

347. The ground state electronic configuration of C_2 using all electrons is

- (A) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \sigma_{2p}^2$
 (B) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \sigma_{2p}^{*2}$
 (C) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^2 \sigma_{2p}^1 \sigma_{2p}^{*1}$
 (D) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^4$

348. Match the following items of column I with the appropriate items in column II.

Column I Molecular species property		Column II Bond order and magnetic	
P	O_2^-	I	2.5 paramagnetic
Q	O_2	II	2.0 diamagnetic
R	O_2^+	III	1.5 diamagnetic
S	O_2^{-2}	IV	1.0 diamagnetic
		V	2.0 paramagnetic
		VI	1.5 paramagnetic

- (A) P-III, Q-V, R-IV, S-III
 (B) P-III, Q-II, R-I, S-IV
 (C) P-III, Q-V, R-IV, S-VI
 (D) P-VI, Q-V, R-I, S-IV

349. For a homonuclear diatomic molecule, the bonding molecular orbital is

- (A) σ_u of lowest energy
 (B) σ_u of second lowest energy
 (C) π_g of lowest energy
 (D) π_u of lowest energy

350. The highest occupied molecular orbital of HF is

- (A) bonding
 (B) antibonding
 (C) ionic
 (D) nonbonding

351. The bond order of C_2 molecule is

- (A) 0 (B) 1
 (C) 2 (D) 3

352. Which of the following has same bond order as NO^+ has?

- (A) CN^- (B) O_2^-
 (C) CN^+ (D) None of them

353. Which of the following statements are correct?

- I. Bond order of NO is 2.5
 II. Bond order of NO^+ is 3.0
 III. Bond order of O_2 is 1.5
 IV. Bond order of CO is 3.0

- (A) I, II, III
 (B) I, II, IV
 (C) I, III, IV
 (D) II, III, IV

354. According to MO theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ ?

- (A) Paramagnetic and bond order $< O_2$.
 (B) Paramagnetic and bond order $> O_2$.
 (C) Diamagnetic and bond order $< O_2$.
 (D) Diamagnetic and bond order $> O_2$.

355. The correct order in which the O–O bond length increases in the following is

- (A) $O_2 < O_3 < H_2O_2$ (B) $H_2O_2 < O_3 < O_2$
 (C) $O_3 < O_2 < H_2O_2$ (D) $O_2 < H_2O_2 < O_3$

356. Which one of the following species is paramagnetic?

- (A) O_2^- (B) CN^-
 (C) CO (D) NO^+

357. Molecular orbital electronic configuration for X_2^{n-} anion is

$$KK^* (\sigma 2s)^2 (\sigma 2s)^2 (\pi 2p_x)^2 (\pi 2p_y)^2 (\sigma 2p_z)^2 (\pi 2p_x)^1$$

The X_2^{n-} is:

- (A) N_2^- (B) O_2^-
 (C) N_2^{2-} (D) O_2^{2-}

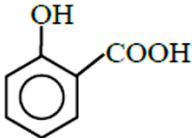
- 358.** Which of the following MO has two nodal planes perpendicular to each other
 (A) σ_{2s} (B) $\pi 2p_x$
 (C) $\pi^* 2p_y$ (D) $\sigma^* 2p_z$
- 359.** Which pair among the following has identical bond order?
 (A) CN^- , CO^+ (B) O_2^- , O_2^+
 (C) CN^- , CN^+ (D) NO^+ , CN^-
- 360.** The correct increasing order of O–O bond length in the following molecules is
 (A) $\text{H}_2\text{O}_2 < \text{O}_2 < \text{O}_3$
 (B) $\text{O}_3 < \text{H}_2\text{O}_2 < \text{O}_2$
 (C) $\text{O}_2 < \text{H}_2\text{O}_2 < \text{O}_3$
 (D) $\text{O}_2 < \text{O}_3 < \text{H}_2\text{O}_2$
- 361.** In the formation of N_2^+ from N_2 , the electron is lost from a:
 (A) σ -orbital
 (B) π -orbital
 (C) σ^* -orbital
 (D) π^* -orbital
- 362.** Which of the following species is paramagnetic?
 (A) O_2^- (B) CN^-
 (C) CO (D) NO^+
- 363.** The bond order of a molecule is given by
 (A) The difference between the number of electrons in bonding and antibonding orbitals
 (B) Total number of electrons in bonding and antibonding orbitals
 (C) Twice the difference between the number of electrons in bonding and antibonding orbitals
 (D) Half the difference between number of electrons in bonding and antibonding orbitals.
- 364.** In a homonuclear molecule which of the following set of orbitals are degenerate?
 (A) $\sigma 1s$ and $\sigma 2s$
 (B) $\pi 2p_x$ and $\pi 2p_y$
 (C) $\pi 2p_x$ and $\sigma 2p_z$
 (D) $\sigma 2p_x$ and $\sigma^* 2p_z$
- 365.** Which of the following is non-existent according to molecular orbital theory?
 (A) H_2^- (B) O_2^-
 (C) He_2 (D) O_2^+
- 366.** The calculated bond order in H_2^- ion is
 (A) 0 (B) $\frac{1}{2}$
 (C) $-\frac{1}{2}$ (D) 1
- 367.** Which of the following representations of wave functions of molecular orbitals and atomic orbitals is not correct?
 (A) $\psi_{\text{MO}} = \psi_A \pm \psi_B$ (B) $\sigma = \psi_A + \psi_B$
 (C) $\sigma^* = \psi_A - \psi_B$ (D) $\psi_{\text{MO}} = \psi_A \times \psi_B$
- 368.** The electronic configuration of carbon is $1s^2 2s^2 2p^2$. There are 12 electrons in C_2 . The correct electronic configuration of C_2 molecule is
 (A) $(\sigma 1s^2)(\sigma^* 1s^2)(\sigma 2s^2)(\sigma^* 2s^2)(\sigma 2p_z^2)(\pi 2p_x^2)$
 (B) $(\sigma 1s^2)(\sigma^* 1s^2)(\sigma 2s^2)(\sigma^* 2s^2)(\pi 2p_x^2 = \pi 2p_y^2)$
 (C) $(\sigma 1s^2)(\sigma^* 1s^2)(\sigma 2s^2)(\sigma^* 2s^2)(\sigma 2p_z^2)(\pi 2p_x^1 = \pi 2p_y^1)$
 (D) $(\sigma 1s^2)(\sigma^* 1s^2)(\sigma 2s^2)(\sigma^* 2s^2)(\pi 2p_x^2 = \pi 2p_y^1)$
- 369.** Which of the following molecules is paramagnetic in nature?
 (A) H_2 (B) Li_2
 (C) B_2 (D) N_2
- 370.** O_2 molecule is
 (A) Paramagnetic (B) Diamagnetic
 (C) Ferromagnetic (D) None of these

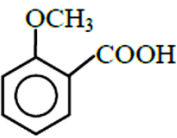
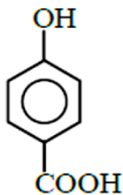
- 371.** From elementary molecular orbital we can deduce the electronic configuration of the singly positive nitrogen molecular ion N_2^+ as
 (A) $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \pi 2p^4 \sigma 2p^1$
 (B) $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p^2 \pi 2p^2 \pi^* 2p^1$
 (C) $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p^2 \pi 2p^4$
 (D) $\sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 \sigma 2p^2 \pi 2p^2$
- 372.** N_2 and O_2 are converted to monocations N_2^+ and O_2^+ . Which of the following is wrong
 (A) In N_2^+ , the N–N bond weakens
 (B) In O_2^+ , the O–O bond order increases
 (C) In O_2^+ , the paramagnetism decreases
 (D) N_2^+ becomes diamagnetic
- 373.** H.O.M.O. (Highest Occupied Molecular Orbital) of CO molecule is
 (A) Non-bonding M.O. with slight antibonding character
 (B) Non-bonding M.O. with slight bonding character
 (C) Pure non-bonding M.O.
 (D) None of the above
- 374.** Which of the following relationship is true?
 (A) Bond dissociation energy of O_2 and O_2^- are same
 (B) Bond dissociation energy of O_2^+ is higher than O_2
 (C) Bond dissociation energy of O_2^- and O_2^{2-} are same
 (D) Bond dissociation energy of O_2^{2-} is higher than O_2^-
- 375.** The bond order in CO_3^{2-} ion between C–O is
 (A) Zero
 (B) 0.88
 (C) 1.33
 (D) 2
- 376.** During the formation of a molecular orbital from atomic orbitals of the same atom, probability of electron density is
 (A) non-zero in the nodal plane
 (B) maximum in the nodal plane
 (C) zero in the nodal plane
 (D) zero on the surface of the lobe
- 377.** Which one of the following cannot exist on the basis of molecular orbital theory?
 (A) H_2^+ (B) He_2^+
 (C) C_2 (D) He_2
- 378.** Which of the following has fractional bond order?
 (A) B_2 (B) O_2^{2-}
 (C) F_2 (D) H_2^-
- 379.** In which of the following set, the values of bond orders will be 2.5?
 (A) O_2^+ , NO , NO^{2+} , CN
 (B) CN , NO^{2+} , CN^- , F_2
 (C) O_2^+ , NO^{2+} , CN^- , F_2
 (D) O_2^+ , NO^{2+} , O_2^{2+} , CN^-
- 380.** Among the following species, which has the minimum bond length?
 (A) B_2 (B) C_2
 (C) F_2 (D) O_2^-
- 381.** Number of antibonding electrons in N_2 is
 (A) 4 (B) 10
 (C) 12 (D) 14
- 382.** Pick out the incorrect statement.
 (A) N_2 has greater dissociation energy than N_2^+
 (B) O_2 has lower dissociation energy than O_2^+
 (C) Bond length in N_2^+ is less than N_2
 (D) Bond length in NO^+ is less than in NO

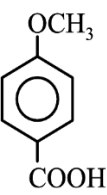
- 383.** Which of the following pairs have identical values of bond order?
 (A) N_2^+ and O_2^+ (B) F_2 and Ne_2
 (C) O_2 and B_2 (D) C_2 and N_2
- 384.** A simplified application of MO theory to the hypothetical 'molecule' OF would give its bond order as
 (A) 2 (B) 1.5
 (C) 1.0 (D) 0.5
- 385.** Which of the following species is paramagnetic?
 (A) NO^- (B) O_2^{2-}
 (C) CN^- (D) CO
- 386.** The following molecules/species have been arranged in the order of their increasing bond orders Identify the correct order.
 (I) O_2 (II) O_2^-
 (III) O_2^{2-} (IV) O_2^+
 (A) $\text{III} < \text{II} < \text{I} < \text{IV}$
 (B) $\text{IV} < \text{III} < \text{II} < \text{I}$
 (C) $\text{III} < \text{II} < \text{IV} < \text{I}$
 (D) $\text{II} < \text{III} < \text{I} < \text{IV}$
- 387.** Which the following molecules / species have identical bond order and same magnetic properties?
 (I) O_2 (II) NO
 (III) N_2^+
 (A) (I), (II) only (B) (I) and (III) only
 (C) (I), (II) and (III) (D) (II) and (III) only
- 388.** Negative bond order means
 (A) molecule is unstable
 (B) molecule is stable
 (C) molecule is neutral
 (D) none of them
- 389.** Which of the following is correct order of σ -bond strength?
 (I) $2s-2s$ (II) $2s-2p$
 (III) $2p-2p$ (IV) $3s-3s$
 (A) $\text{I} > \text{II} > \text{III} > \text{IV}$ (B) $\text{III} > \text{II} > \text{I} > \text{IV}$
 (C) $\text{IV} > \text{I} > \text{II} > \text{III}$ (D) $\text{III} > \text{I} > \text{II} > \text{IV}$
- 390.** Both $\text{N}(\text{SiH}_3)_3$ and $\text{NH}(\text{SiH}_3)_2$ compounds have trigonal planar skeleton. Incorrect statement about both compounds is:
 (A) $\text{Si}\hat{\text{N}}\text{Si}$ bond angle in $\text{NH}(\text{SiH}_3)_2 > \text{Si}\hat{\text{N}}\text{Si}$ bond angle in $\text{N}(\text{SiH}_3)_3$
 (B) $\text{N}-\text{Si}$ bond length in $\text{NH}(\text{SiH}_3)_2 > \text{N}-\text{Si}$ bond length in $\text{N}(\text{SiH}_3)_3$
 (C) $\text{N}-\text{Si}$ bond length in $\text{NH}(\text{SiH}_3)_2 < \text{N}-\text{Si}$ bond length in $\text{N}(\text{SiH}_3)_3$
 (D) Back bonding strength in $\text{NH}(\text{SiH}_3)_2 > \text{Back bonding strength in } \text{N}(\text{SiH}_3)_3$
- 391.** The incorrect statement regarding $\text{O}(\text{SiH}_3)_2$ and OCl_2 molecule is/are:
 (A) The strength of back bonding is more in $\text{O}(\text{SiH}_3)_2$ molecule than OCl_2 molecule
 (B) $\text{Si}-\hat{\text{O}}-\text{Si}$ bond angle in $\text{O}(\text{SiH}_3)_2$ is greater than $\text{Cl}-\hat{\text{O}}-\text{Cl}$ bond angle in OCl_2
 (C) The nature of back bond in both molecules is $2p\pi-3d\pi$
 (D) Hybridisation of central O-atom in both molecules is same
- 392.** Among following molecule $\text{N}-\text{Si}$ bond length is shortest:
 (A) $\text{N}(\text{SiH}_3)_3$
 (B) $\text{NH}(\text{SiH}_3)_2$
 (C) $\text{NH}_2(\text{SiH}_3)$
 (D) All have equal $\text{N}-\text{Si}$ bond length
- 393.** Which of the following molecule has weakest ($p\pi-d\pi$) back bonding?
 (A) OCl_2 (B) $\text{N}(\text{SiH}_3)_3$
 (C) SiF_4 (D) $\text{O}(\text{SiH}_3)_2$

- 394.** "Hybridisation of central atom does not always change due to back bonding". This statement is valid for which of the following compounds?
 (i) CCl_3^- (ii) CCl_2
 (iii) $(\text{SiH}_3)_2\text{O}$ (iv) $\text{N}(\text{SiH}_3)_3$
 (A) (i), (ii) (B) (i), (iii)
 (C) (ii), (iii) (D) All
- 395.** The shape with respect to the central atom of the following molecules are:
 $\text{N}(\text{SiH}_3)_3$; $\text{Me}_3\text{N}; (\text{SiH}_3)_3\text{P}$
 (A) planar, pyramidal, planar
 (B) planar, pyramidal, pyramidal
 (C) pyramidal, pyramidal, pyramidal
 (D) pyramidal, planar, pyramidal
- 396.** Incorrect statement regarding BF_2NH_2 molecule is:
 (A) FBF bond angle $< 120^\circ$
 (B) HNH bond angle $> 109^\circ 28'$
 (C) Exhibits intermolecular H-bond
 (D) Hybridization of N -atom is sp^3
- 397.** The number of antibonding electron pairs in O_2^{2-} molecular ion on the basis of molecular orbital theory is (at no. $\text{O} = 8$):
 (A) 2 (B) 3
 (C) 4 (D) 5
- 398.** Which of the following option w.r.t. increasing bond order is correct?
 (A) $\text{NO} < \text{C}_2 < \text{O}_2^- < \text{H}_2^+$
 (B) $\text{C}_2 < \text{NO} < \text{H}_2^+ < \text{O}_2^-$
 (C) $\text{H}_2^+ < \text{O}_2^- < \text{NO} < \text{C}_2$
 (D) $\text{H}_2^+ < \text{O}_2^- < \text{C}_2 < \text{NO}$
- 399.** The bond-orders of the given species are such that-
 (A) $\text{O}_2^- > \text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2$
 (B) $\text{O}_2^+ > \text{O}_2 > \text{O}_2^- > \text{O}_2^{2-}$
 (C) $\text{O}_2^+ > \text{O}_2^{2-} > \text{O}_2^- > \text{O}_2$
 (D) $\text{O}_2^{2-} > \text{O}_2 > \text{O}_2^+ > \text{O}_2^-$
- 400.** The ion that is isoelectronic with CO and having same bond order is :
 (A) CN^- (B) O_2^+
 (C) O_2^- (D) N_2^+
- 401.** Which of the following is paramagnetic:
 (A) O_2^- (B) CN^-
 (C) CO (D) NO^+
- 402.** In the following which of the two are paramagnetic:
 (a) N_2 (b) CO
 (c) B_2 (d) NO_2
 Correct answer is:
 (A) a & c (B) b & c
 (C) c & d (D) b & d
- 403.** Increasing order of bond length in NO, NO^+ and NO^- is:
 (A) $\text{NO} > \text{NO}^- > \text{NO}^+$
 (B) $\text{NO}^+ < \text{NO} < \text{NO}^-$
 (C) $\text{NO} < \text{NO}^+ < \text{NO}^-$
 (D) $\text{NO} < \text{NO}^+ = \text{NO}^-$
- 404.** The paramagnetic property of oxygen is well explained by:
 (A) Molecular orbital theory
 (B) Resonance theory
 (C) Valence bond theory
 (D) VSEPR theory
- 405.** Which of the following has fractional bond order?
 (A) O_2^{2+} (B) O_2^{2-}
 (C) F_2^{2-} (D) H_2^-
- 406.** Higher is the bond order, greater is:
 (A) Bond dissociation energy
 (B) Covalent character
 (C) Bond length
 (D) Paramagnetism

- 407.** The bond order of CO molecule on the basis of molecular orbital theory is:
 (A) Zero (B) 2
 (C) 3 (D) 1
- 408.** Which of the following group of molecules have $2\frac{1}{2}$ bond order:
 (A) $\text{N}_2^{-2}, \text{O}_2^{-2}, \text{CO}$ (B) $\text{N}_2^{+}, \text{O}_2^{+}, \text{NO}$
 (C) $\text{C}_2^{-2}, \text{BN}, \text{O}_2$ (D) $\text{CN}^{-}, \text{NO}^{+}, \text{O}_2^{-2}$
- 409.** Which of the following species will have the minimum bond energy:
 (A) N_2 (B) N_2^{-}
 (C) N_2^{+} (D) N_2^{-2}
- 410.** Which of the following statements is true?
 (A) HF is less polar than HBr
 (B) Water does not contain any ions
 (C) Chemical bond formation takes place when forces of attraction overcome the forces of repulsion
 (D) In covalent cond, transfer of electrons takes place
- 411.** A square planar complex is formed by hybridisation of which atomic orbital?
 (A) s, p_x, p_y, d_{yz} (B) $s, p_x, p_y, d_{x^2-y^2}$
 (C) s, p_x, p_y (D) s, p_x, p_y, d_{xy}
- Hydrogen Bond: Types of Hydrogen Bonds**
- 412.** Which concept best explains that o-nitrophenol is more volatile than p-nitrophenol?
 (A) Resonance (B) Steric hinderance
 (C) Hydrogen bond (D) Hyperconjugation
- 413.** In which of the following hydrogen bond is present
 (A) H_2 (B) Ice
 (C) Sulphur (D) Hydrocarbon
- 414.** In the following which has highest boiling point
 (A) HI (B) HF
 (C) HBr (D) HCl
- 415.** Which contains hydrogen bond
 (A) HF (B) HCl
 (C) HBr (D) HI
- 416.** Contrary to other hydrogen halides, hydrogen fluoride is a liquid because
 (A) Size of F atom is small
 (B) HF is a weak acid
 (C) HF molecule are hydrogen bonded
 (D) Fluorine is highly reactive
- 417.** Which contains strongest H-bond
 (A) $\text{O}-\text{H}\cdots\text{S}$ (B) $\text{S}-\text{H}\cdots\text{O}$
 (C) $\text{F}-\text{H}\cdots\text{F}$ (D) $\text{F}-\text{H}\cdots\text{O}$
- 418.** The hydrogen bond is strongest in
 (A) Water (B) Ammonia
 (C) Hydrogen fluoride (D) Acetic acid
- 419.** Strength of hydrogen bond is intermediate between
 (A) Vander Waal and covalent
 (B) Ionic and covalent
 (C) Ionic and metallic
 (D) Metallic and covalent
- 420.** Which of the following compounds has the highest boiling point
 (A) HCl (B) HBr
 (C) H_2SO_4 (D) HNO_3
- 421.** The hydrogen bond is strongest in:
 (A) $\text{O}-\text{H}\cdots\text{S}$ (B) $\text{S}-\text{H}\cdots\text{O}$
 (C) $\text{F}-\text{H}\cdots\text{F}$ (D) $\text{O}-\text{H}\cdots\text{O}$
- 422.** H_2O boils at higher temperature than H_2S , because it is capable of forming:
 (A) Ionic bonds
 (B) Covalent bonds
 (C) Hydrogen bonds
 (D) Metallic bonds

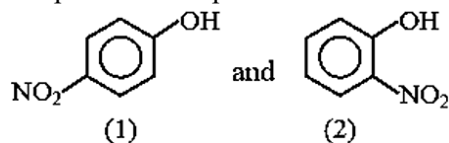
423. The correct order of volatility is :
 (A) $\text{NH}_3 < \text{H}_2\text{O}$
 (B) p-nitro phenol < o-nitro phenol
 (C) $\text{CH}_3\text{OH} > \text{CH}_3 - \text{O} - \text{CH}_3$
 (D) $\text{HF} > \text{HCl}$
424. Intra molecular H-bond:
 (A) Decreases Volatility
 (B) Increases melting point
 (C) Increases viscosity
 (D) Increases vapour pressure
425. Weakest hydrogen bond is:
 (A) $\text{O}-\text{H} \cdots \cdots \text{N}$ (B) $\text{S}-\text{H} \cdots \cdots \text{S}$
 (C) $\text{F}-\text{H} \cdots \cdots \text{F}$ (D) $\text{N}-\text{H} \cdots \cdots \text{N}$
426. Which of the following has strongest intra molecular hydrogen bonding:
- (A) 

(B) 
- (C) 

(D) 
427. Acetic acid exists as dimer in benzene due to:
 (A) Condensation reaction
 (B) Hydrogen bonding
 (C) Presence of carboxyl group
 (D) None of the above
428. In which case hydrogen bond will not be observed-
 (A) H_3O_2^- (B) H_2O
 (C) HF (D) H_3O^+
429. The high boiling point of water is due to the presence of:
 (A) Dative bond
 (B) Covalent bond
 (C) Hydrogen bond
 (D) Vander Waals bond
430. Maximum no. of hydrogen bonds formed by a water molecule in ice is:
 (A) 4 (B) 3
 (C) 2 (D) 1
431. Hydrogen bonding is formed in compounds containing hydrogen and:
 (A) Highly electro-negative atoms
 (B) Highly electro-positive atoms
 (C) Metal atoms with d-orbitals occupies
 (D) Metalloids
432. The intermolecular force in hydrogen fluoride is due to:
 (A) Dipole induced dipole interactions
 (B) Ion-dipole interactions
 (C) Hydrogen bond
 (D) None of these
433. Density of ice is less than that of water because of:
 (A) presence of hydrogen bonding
 (B) crystal modification of ice
 (C) open porous structure of ice due to hydrogen bonding
 (D) different physical states of these
434. Which of the following compounds show intramolecular hydrogen bonding:
 (A) o-nitrophenol (B) p-nitrophenol
 (C) phenol (D) salicyl aldehyde
435. The pair of molecules forming strongest hydrogen bonds are:
 (A) SiH_4 and SiF_6
 (B) $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{CH}_3$ and CHCl_3
 (C) $\text{H} - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH}$ and $\text{CH}_3 - \overset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{OH}$
 (D) H_2O and H_2
436. The boiling point of a compound is raised by:
 (A) intermolecular hydrogen bonding
 (B) high volatility
 (C) intramolecular hydrogen bonding
 (D) non-polarity

437. The boiling point of methanol is greater than methyl thiol because:
- There is intramolecular hydrogen bonding in methanol and intermolecular hydrogen bonding in methyl thiol
 - There is intermolecular hydrogen bonding in methanol and no hydrogen bonding in methyl thiol
 - There is no hydrogen bonding in methanol and intermolecular hydrogen bonding in methyl thiol
 - There is intramolecular hydrogen bonding in methanol and no hydrogen bonding in methyl thiol

438. Out of the two compounds shown below, the vapour pressure of (2) at a particular temperature is expected to be:



- Higher than that of (1)
 - Lower than that of (1)
 - Same as that of (1)
 - Can be higher or lower depending upon the size of the vessel
439. The strongest P–O bond is found in the molecule
- F₃PO
 - Cl₃PO
 - Br₃PO
 - (CH₃)₃PO
440. Consider the following compounds:
- (I) ClF₃ (II) BrF₃
- The order of the angles between axial and equatorial bond pairs is
- I > II
 - I < II
 - I = II
 - None of these
441. Out of C₂H₆, C₂H₄ and C₂H₂. Compound which have highest C–C bond length is:
- C₂H₄
 - C₂H₂
 - C₂H₆
 - All have equal C–C bond length

442. Which of the following molecule has 3c – 4e[–] bond?
- Al₂Cl₆
 - Be₂Cl₄
 - I₂Cl₆
 - All are having 3c - 4e- bond

443. No X–X bond exists in which of the following compounds having general form of X₂H₆?
- B₂H₆
 - C₂H₆
 - Al₂H₆
 - Si₂H₆

444. Which of the following molecules have Correct indicated overlapping.

Molecule	Overlapping in the bridge bond (if any)
(A) Si ₂ Cl ₆	sp ³ – p – sp ³
(B) Be ₂ H ₄	sp ² – s – sp ²
(C) Si ₂ H ₆	sp ³ – s – sp ³
(D) B ₂ H ₆	sp ³ – s – sp ³

445. Which of the following has highest bond energy?

(A) C–F(CF ₄)	(B) C–Cl(CCl ₄)
(C) C–Br(CBr ₄)	(D) B–F(BF ₃)

446. Choose the correct angle order.

(A) HPH in PH ₄ ⁺ = HCH in CH ₄
(B) HNH in NH ₃ < HPH in PH ₃
(C) HNH in NH ₃ < HPH in PH ₄ ⁺
(D) OSO ₃ SO ₃ ^{2–} < ONO ₃ NO ₃ [–]

447. In which of the following back bonding is not possible-

(A) N(CH ₃) ₃	(B) BO ₃ ^{3–}
(C) P(CH ₃) ₃	(D) BF ₄ [–]

448. Molecule(s) in which number of atoms in a plane may be ten.

(A) B ₂ Me ₄ H ₂	(B) Al ₂ (CH ₃) ₆
(C) N(SiH ₃) ₃	(D) [Co(NH ₃) ₆] ³⁺

449. Which of the following statement(s) is/are correct-

- (A) $[\text{B}_3\text{O}_6]^{-3}$ ion is non-planar but $\text{B}_3\text{N}_3\text{H}_6$ is planar
 (B) $(\text{H}_3\text{Si})_3\text{N}$ is planar but $(\text{SiH}_3)_3\text{P}$ is pyramidal
 (C) $\text{H}_3\text{C}-\text{NCS}$ is bent but SiH_3-NCS is linear
 (D) $(\text{CH}_3)_3\text{N}$ is pyramidal but $(\text{GeH}_3)_3\text{N}$ is planar

450. Which of the following order is correct for covalent character (if the cation sizes are identical in pairs)-

- (A) $\text{NaCl} < \text{CuCl}$ (B) $\text{CaCl}_2 < \text{PdCl}_2$
 (C) $\text{RbCl} < \text{AuCl}$ (D) All of these

451. Select the correct order of acidic nature-

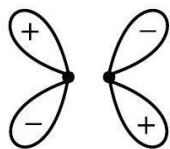
- (A) $\text{CO} > \text{CO}_2$ (B) $\text{SiO}_2 < \text{CO}_2$
 (C) $\text{NO}_2 > \text{N}_2\text{O}_5$ (D) $\text{SO}_2 > \text{SO}_3$

452. Which of the following order is correct-

- (A) $\text{Li}_{(\text{aq})}^+ > \text{Na}_{(\text{aq})}^+ > \text{K}_{(\text{aq})}^+$; Ionic mobility
 (B) $\text{Li}_{(\text{aq})}^+ < \text{Na}_{(\text{aq})}^+ < \text{K}_{(\text{aq})}^+$; Hydrated size
 (C) $\text{F}_{(\text{aq})}^- > \text{Cl}_{(\text{aq})}^- > \text{Br}_{(\text{aq})}^-$; Hydrated size
 (D) $\text{F}_{(\text{aq})}^- > \text{Cl}_{(\text{aq})}^- > \text{Br}_{(\text{aq})}^-$; Ionic mobility

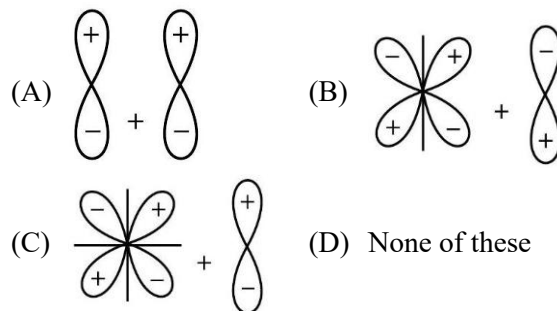
453. Choose the correct statement for O_2 molecule. As per M.O.T. both the unpaired electrons of O_2 molecule are

- (A) having same energy
 (B) in molecular orbitals which have given shape:



- (C) in molecular orbitals which are perpendicular to each other
 (D) All are correct

454. Which of the following orbital overlapping results antibonding, gerade M.O. having two nodal plane:



455. Which of the following statements is incorrect-

- (A) The electron is being added to π^* orbital to get CO^- from CO
 (B) The electron is being removed from π^* orbital to get CO^+ from CO
 (C) In the formation of CO^- from CO , bond length increases
 (D) In the formation of CO^+ from CO , bond length decreases

456. Electronic configuration of H_2^- is $\sigma_{1s}^2, \sigma_{1s}^{*1}$ but not $\sigma_{1s}^3, \sigma_{1s}^{*0}$ because-

- (A) Hund's rule is violated
 (B) Bond order becomes zero
 (C) Pauli exclusion principle is violated
 (D) Fractional bond order is not possible

457. Select the species in which cation is diamagnetic but anion is paramagnetic-

- (A) $\text{NO}[\text{BF}_4]$ (B) $\text{O}_2[\text{AsF}_6]$
 (C) BaO_2 (D) KO_2

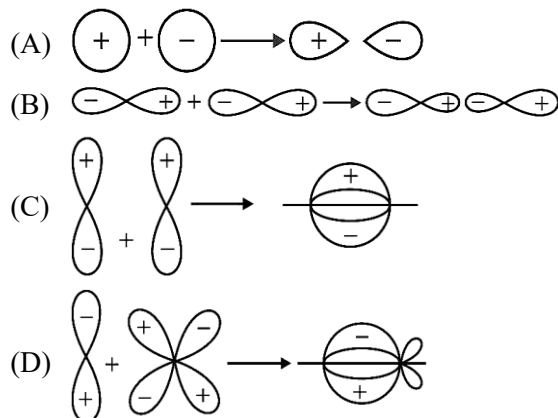
458. The molecule/specie does not exist

- (A) N_2^+ (B) He_2^+
 (C) O_2^{2-} (D) Be_2

459. In which of the following process the increase in bond length is highest-

- (A) $\text{N}_2 \longrightarrow \text{N}_2^+$ (B) $\text{N}_2 \longrightarrow \text{N}_2^-$
 (C) $\text{N}_2^+ \longrightarrow \text{N}_2^-$ (D) $\text{N}_2^- \longrightarrow \text{N}_2^+$

460. Which of the following molecular orbital formation is incorrect.



461. O_2 is paramagnetic, this property as for B_2 , can not be explained by the traditional Lewis dot structure but is evident from the MO picture. In energy level diagram of O_2^{2+} , 13th e^- present in

- (A) σ_{2p_z} orbital
 (B) π_{2p_y} or π_{2p_x} orbital
 (C) $\pi_{2p_y}^*$ or $\pi_{2p_x}^*$ orbital
 (D) $\sigma_{2p_z}^*$ orbital

462. Which of the following set of orbital is results, N.B.M.O [If z-axis is the bond formation axis]

- (A) $p_z + d_{xz}$
 (B) $s + d_{yz}$
 (C) $s + d_{xz}$
 (D) All of these

463. Which of following set of molecules/ion has equal bond order-

- (A) NO^+ , N_2 , O_2^{2+}
 (B) O_2 , N_2^{2-} , NO^-
 (C) NO , O_2^+ , N_2^-
 (D) (A), (B) and (C)

464. Choose the correct ionization energy order from the following options-

- (A) $F_2 > F$ (B) $B > B_2$
 (C) $N_2 < N$ (D) $B_2 > B$

465. Which of the following orbital is having maximum number of nodal plane-

- (A) σ_{p-p}
 (B) π_{d-d} (2 lobes interaction)
 (C) π_{p-p}^*
 (D) π_{d-p}

466. In the process of $N_2^- \rightarrow N_2$ the electron is removed from ' ____ ' provided the x -axis is the bond formation axis.

- (A) $\pi_{2p_y}^*$ orbital (B) $\sigma_{2p_x}^*$ orbital
 (C) σ_{2p_x} orbital (D) σ_{2s} orbital

467. The species having co-ordinate bond from the following option is/are.

- (A) CO_3^{2-} (B) NO_3^-
 (C) NO_2^- (D) BF_4^-

468. Which of the following is an example of super octet molecule?

- (A) ClF_3 (B) PCl_5
 (C) IF_7 (D) All the three

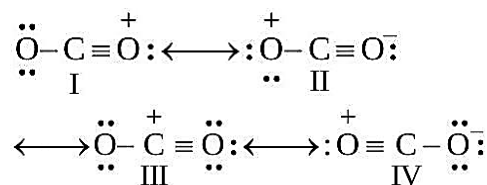
469. A covalent bond is possible between

- (A) Only similar atoms
 (B) Only dissimilar atoms
 (C) Similar and dissimilar atoms
 (D) Similar molecules

470. Which of the following ions has resonating structure?

- (A) SO_4^{2-} (B) PO_4^{3-}
 (C) SO_3^{2-} (D) All of these

471. Which of the following resonating structure is correct for CO_2 ?



- (A) I (B) II
 (C) III (D) IV

472. PCl_5 is formed but NCl_5 does not why?
473. SF_4 is formed but OF_4 does not why?
474. Give the average formal charge and average bond order of X–O bond in the following compounds?
- (i) SO_4^{2-} (ii) SO_3^{2-}
 (iii) ClO_4^- (iv) PO_4^{3-}
475. In which molecule s-p overlapping occurs?
- (A) CH_4 (B) NH_3
 (C) H_2O (D) All are wrong
476. The strongest covalent bond is formed by the overlap of
- (A) 2 s and 2 p orbitals
 (B) 3 s and 3 s orbitals
 (C) 3p and 3d orbitals
 (D) 3p and 3p collateral orbitals
477. Which of the following orbitals can be formed σ bond?
- (A) s-orbital (B) p-orbital
 (C) d-orbital (D) None of these
478. Which of the following compound is formed in the second excitation state of sulphur atom?
- (A) SF_4 (B) SF_6
 (C) SF_2 (D) None of these
479. Fluorine molecule is formed by:
- (A) The axial p-p orbital overlap
 (B) The sideways p–p orbital overlap
 (C) The s-s orbital overlap
 (D) The s-p orbital overlap
480. Compound in which coordinate covalent π bond is present:
- (A) NO_3^- (B) SF_6
 (C) CO (D) HNC
481. Molecule in which bond order is > 2
- (A) CO_3^{2-} (B) NH_4^+
 (C) N_3^- (D) None of these
482. Which of the following statements is correct?
- (A) In CO carbon has positive formal charge.
 (B) In CO_2 resonance is absent.
 (C) In COCl_2 bond order 1.33
 (D) In NOF_3 coordinate covalent bond is present.
483. What are the total number of σ & π bonds in tetracyanomethane.
484. O_2 exists but S_2 does not exist at room temperature why?
485. Select pair of compounds in which both have different hybridisation but have same molecular geometry.
- (A) $\text{BF}_3, \text{BrF}_3$ (B) $\text{ICl}_2^-, \text{BeCl}_2$
 (C) $\text{BCl}_3, \text{PCl}_3$ (D) $\text{PCl}_3, \text{NCl}_3$
486. Choose the correct option for following statements:
- (I) sp^3 hybrid orbitals are at 90° to one another.
 (II) sp^3d^2 adjacent hybrid orbitals are at 90° to one another.
 (III) sp^2 hybrid orbitals are at 120° to one another.
 (IV) In $\text{sp}^3\text{d}, \text{sp}^3\text{d}^2$ and sp^3d^3 axial and equatorial positions are present.
- (A) FFTF (B) FTFF
 (C) FTTF (D) FTTT
487. Which of the following specie has sp^3d^3 hybridisation?
- (A) XeF_5^- (B) XeF_6
 (C) XeO_6^{4-} (D) XeF_2
488. All fluorine atoms are in same plane in:
- (A) CHF_3 (B) ClF_3
 (C) XeOF_4 (D) IF_5

- 489.** Select correct statement for BrF_5 .
- (A) All fluorine atoms are in same plane
 (B) Four fluorine atoms and Br atom are in same plane
 (C) Four fluorine atoms are in same plane
 (D) It has all $\text{F}-\text{Br}-\text{F}$ bond angles at 90°
- 490.** In which of following cases, the central atom is not perfectly sp^3 hybridised?
- (A) BF_4^- (B) SiF_4
 (C) CHF_3 (D) CCl_4
- 491.** Molecule which does not contain any $\text{F}-\text{X}-\text{F}$ bond angle which is less than 90° : (X = central atom)
- (A) IF_7 (B) BrF_3
 (C) PF_5 (D) SF_4
- 492.** The $\text{H}-\text{O}-\text{H}$ bond angles in H_3O^+ are approximately 107° . The orbitals used by oxygen in these bonds are best described as:
- (A) p-orbitals
 (B) sp-hybrid orbitals
 (C) sp^2 -hybrid orbitals
 (D) sp^3 -hybrid orbitals
- 493.** Which of the following shape are not possible for possible value of 'n' in XeF_n
- (A) Linear
 (B) Square planar
 (C) Trigonal planar
 (D) Capped octahedral
- 494.** The electron pair geometry of NH_3 is
- (A) Pyramidal (B) Tetrahedral
 (C) Angular (D) V-shape
- 495.** In BClBrI molecule the maximum % s-character provided from the central atom is in bond:
- (A) B-I (B) B-Cl
 (C) B-Br (D) Cannot predict
- 496.** The maximum % of s-character in $\text{N}-\text{H}$ bond is observed in
- (A) NH_3 (B) NH_4^+
 (C) N_2H_4 (D) N_2H_2
- 497.** Maximum seven atoms are present in a plane, Except in
- (A) H_2CSF_4
 (B) $\text{P}(\text{CH}_3)_2(\text{CH}_3)_3$
 (C) $\text{N}(\text{SiH}_3)_3$
 (D) $\text{B}(\text{CH}_3)_3$
- 498.** In which of the following case $\text{C}-\text{C}$ bond length will be highest?
- (A) CF_3-CF_3 (B) $\text{F}_2\text{CH}-\text{CHF}_2$
 (C) $\text{FCH}_2-\text{CH}_2\text{F}$ (D) CH_3-CF_3
- 499.** Select molecule(s) in which all $\text{P}-\text{Cl}$ lengths are identical:
- (A) PCl_3F_2 (B) PCl_3
 (C) PCl_2F_3 (D) None of these
- 500.** Which of the set is incorrect for the following compound of their bond length?
- (A) $d_{\text{C}-\text{Cl}} : \text{CH}_3\text{Cl} < \text{CF}_3\text{Cl}$
 (B) $d_{\text{N}-\text{N}} : \text{N}_2\text{H}_4 > \text{N}_2$ F_4
 (C) $d_{\text{O}-\text{O}} : \text{O}_2 \text{ F}_2 > \text{H}_2\text{O}_2$
 (D) $d_{\text{C}-\text{C}} : \text{C}_2 \text{ F}_6 < \text{C}_2\text{H}_6$
- 501.** Select the correct statement -
- (A) A molecule which have T.B.P. geometry axial bond length always greater than that of equatorial bond length
 (B) In PCl_3F_2 molecule equatorial bond length is greater than that of axial bond length
 (C) A molecule which have P.B.P. geometry, axial orbital length of central atom is greater than that of equatorial orbital length
 (D) A molecule which have P.B.P. geometry, equatorial orbital bond length of central atom is greater than that of axial orbital length

502. According to bent's rule more EN surrounding atom prefer to attach with that hybrid orbital which has less electronegativity.

Select the correct statement for following molecules:

- (i) $\text{PF}_2(\text{CH}_3)_3$ (ii) $\text{PF}_2(\text{CF}_3)_3$
- (A) Both have trigonal bipyramidal structure with respect to P
 (B) P–F bond length is longer in $\text{PF}_2(\text{CH}_3)_3$ than in $\text{PF}_2(\text{CF}_3)_3$
 (C) F-atoms occupy axial position in both
 (D) All of these

503. Maximum number of atom(s) in I_2Cl_6 which is/are in the same plane.

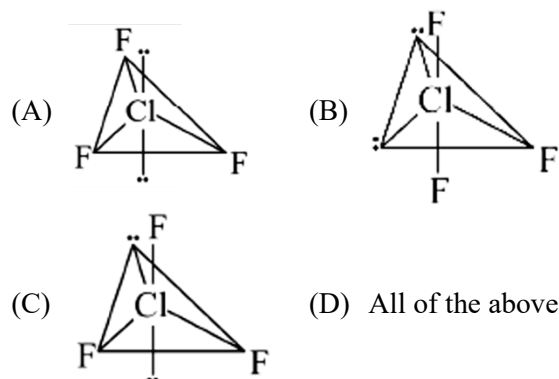
504. Find the odd electron specie in which odd electron is present in pure 'p' atomic orbital.

- (A) $\dot{\text{N}}\text{O}_2$ (B) $\dot{\text{C}}\text{I}\text{O}_2$
 (C) $\dot{\text{C}}\text{F}_3$ (D) $\dot{\text{C}}\text{H}_3$

505. The shape of $[\text{ClF}_4]^-$ and $[\text{ClF}_2]^-$ ions is respectively

- (A) See-saw and linear
 (B) See-saw and bent
 (C) Tetrahedral and linear
 (D) Square planar and linear

506. More correct structure of ClF_3 is:



507. Which of the following molecules/species has the minimum number of lone pairs?

- (A) ICl_3 (B) BF_4^-
 (C) SnCl_2 (D) XeF_4

508. The number of $\hat{\text{F}}\hat{\text{I}}\hat{\text{F}}$ adjacent angles ($90^\circ + 72^\circ$) in IF_7 molecule is

- (A) 10
 (B) 15
 (C) 20
 (D) 14

509. Which of the following is isostructural with XeOF_4 ?

- (A) BrF_5
 (B) PCl_3F_2
 (C) SeF_4
 (D) XeO_2F_2

510. Which of the following is planar?

- (A) XeO_3
 (B) BrF_5
 (C) ClF_3
 (D) All are planar

511. Which of the following molecule contain maximum number of atoms in one plane?

- (A) CCl_4
 (B) PF_5
 (C) XeF_5^-
 (D) SF_6

512. Which of the following statements is incorrect?

- (A) In H_2O , the bond angle is less than OF_2
 (B) In ClF_3 , the axial Cl–F bond length is larger than equatorial Cl–F bond length
 (C) In SF_4 , F–S–F equatorial bond angle is not equal to 120° due to lone pair-bond pair repulsions.
 (D) In ICl_4^- , bond angles is 90°

513. In SNF_3 , the $\angle\text{FSF}$ should be:

- (A) Less than 120° and more than $109^\circ 28'$
 (B) Less than $109^\circ 28'$
 (C) Less than 180° and more than 120°
 (D) Exactly equal to $109^\circ 28'$

- 514.** Give the correct order of initials T or F for following statements. Use T if statement is true and F if it is false:
- (I) The order of repulsion between different pair of electrons is $l_p - l_p > l_p - b_p > b_p - b_p$
- (II) In general, as the number of lone pair of electrons on central atom increases, value of bond angle from normal bond angle also increases
- (III) The number of lone pair on O in H_3O^+ is 2 while on N in NH_3 is 1
- (IV) The electron pair geometry of I_3^- is trigonal bipyramidal.
- (A) T T T F (B) T F F T
(C) T F T T (D) T F F F
- 515.** The number of non-axial set of 'd' orbital participate in the hybridization of the anionic part of PCl_5
- (A) 1 (B) 2
(C) 3 (D) 0
- 516.** The (Cl–P–Cl) adjacent angle difference in cationic part and anionic part in $PCl_5(s)$ is:
- (A) 60° (B) 90°
(C) 19.5° (D) 10.5°
- 517.** Hybridisation of cationic part of $Cl_2O_6(s)$ is:
- (A) sp^2 (B) sp^3
(C) sp^3d (D) sp^3d^2
- 518.** The correct order of bond angle
- (A) $CCl_4 < SiCl_4$
(B) $NO_2^+ > NO_2 > NO_2^-$
(C) $NH_3 = NF_3$
(D) $I_3^+ > I_3^-$
- 519.** Incorrect order of bond length
- (A) N–N bond length : $N_2O_3 > N_2O_4$
(B) N–N bond length : $N_2H_4 < [N_2H_5]^+$
(C) O–F bond length : $O_2F_2 < OF_2$
(D) C–O bond length : $COH_2 < COF_2$
- 520.** Which of the following statement is correct?
- (A) $N(SiH_3)_3$ molecule is planar with respect to N.
(B) In $P(SiH_3)_3$ molecule, P -atom has sp^3 hybridisation.
(C) $N(GeH_3)_3$ has pyramidal shape.
(D) $(GeH_3)NCO$ has linear structure.
- 521.** Correct order of basic strength
- (A) $N(CH_3)_3 > P(SiH_3)_3 > N(SiH_3)_3$
(B) $NH_3 > PH_3$
(C) $H_2O > H_2S$
(D) None of these
- 522.** In $C(CH_3)_4$ molecule, maximum number of atoms are in one plane.
- 523.** Select the correct order of following property.
- (A) % s-character : $sp^3 > sp^2 > sp$
(B) O–N–O bond angle : $NO_3^- > NO_2^+$
(C) All angles in CH_2F_2 are not identical
(D) C–F bond length : $CF_4 > CH_3F > CH_2F_2 > CF_3H$
- 524.** F–As–F bond angle in AsF_2Cl_3 can be
- (A) 90° & 180° only (B) 120° only
(C) 90° , 120° & 180° (D) 90° only
- 525.** Select the incorrect statement(s) about N_2F_4 and N_2H_4 .
- (A) In N_2F_4 , d-orbitals are contracted by electronegative fluorine atoms, but d-orbital contraction is not possible by H - atom in N_2H_4 .
(B) The N–N bond energy in N_2F_4 is more than N–N bond energy in N_2H_4 .
(C) The N–N bond length in N_2F_4 is more than that in N_2H_4 .
(D) The N–N bond length in N_2F_4 is less than that in N_2H_4 .

526. If the %s-character in one Sb–H bond in SbH_3 is 1.0%. What is %p-character in the orbital occupied by its lone pair -

- (A) 99 (B) 97
(C) 90 (D) None of these

527. **Statement-1:** In P.B.P. geometry axial orbitals length are greater than that of equatorial orbital.

Statement-2: IF_7 has P.B.P. geometry.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

528. Find the number of molecule in which hybridisation is not present according to Drago's rule $\text{NH}_3, \text{H}_2\text{S}, \text{H}_2\text{Se}, \text{PH}_3$.

529. In which of the following cases, the strength of back bonding is maximum:

- (A) PF_3 (B) BF_3
(C) $(\text{H}_3\text{Si})_3\text{N}$ (D) $(\text{SiH}_3)_2\text{O}$

530. Choose the incorrect statement from the following-

- (A) Bond angles are not affected in BF_3 due to back bonding
(B) Bond angles are affected in PF_3 due to back bonding
(C) Bond angles are not affected in B(OMe)_3 due to back bonding
(D) None of these

531. **Statement 1:** The B-F bond length in BF_3 is not identical with that in BF_4^- because

Statement 2: Back bonding is involved in BF_4^- but not in BF_3 .

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement- 1 is true, statement- 2 is true and statement- 2 is NOT the correct explanation for statement-1.

(C) Statement- 1 is true, statement- 2 is false.

(D) Statement-1 is false, statement-2 is true.

532. Which of the following statement is/are correct for CCl_3^- and $:\text{CCl}_2$.

- (A) Back bonding in both cases from Cl to C atom
(B) Back bonding in both cases from C to Cl-atom
(C) Back bonding in CCl_3^- from C to Cl but reverse in $:\text{CCl}_2$
(D) Direction of back bonding is just opposite in two cases.

V.B.T., Hybridisation & VSEPR Theory

533. In which of the following pairs the two species are not isostructural?

- (A) AlF_6^{3-} and SF_6 (B) CO_3^{2-} and NO_3^-
(C) PCl_4^+ and SiCl_4 (D) PF_5 and BrF_5

534. In which of the following sets, all the given species are isostructural?

- (A) $\text{BF}_3, \text{NF}_3, \text{PF}_3, \text{AlF}_3$
(B) $\text{PCl}_3, \text{AlCl}_3, \text{BCl}_3, \text{SbCl}_3$
(C) $\text{BF}_4^-, \text{CCl}_4, \text{NH}_4^+, \text{PCl}_4^+$
(D) $\text{CO}_2, \text{NO}_2, \text{ClO}_2, \text{SiO}_2$

535. The formation of molecular complex $\text{BF}_3 - \text{NH}_3$ results in a change in hybridization of boron:

- (A) from sp^3 to sp^3d
(B) from sp^2 to dsp^2
(C) from sp^3 to sp^2
(D) from sp^2 to sp^3

536. The group of molecules having identical shape is:

- (A) $\text{SF}_4, \text{XeF}_4, \text{CCl}_4$
 (B) $\text{ClF}_3, \text{XeOF}_2, \text{XeF}_3^+$
 (C) $\text{PCl}_5, \text{IF}_5, \text{XeO}_2\text{F}_2$
 (D) $\text{BF}_3, \text{PCl}_3, \text{XeO}_3$

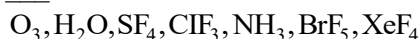
537. The decreasing order of bond angles in $\text{BF}_3, \text{NH}_3, \text{PF}_3$ and I_3^- is:

- (A) $\text{I}_3^- > \text{BF}_3 > \text{NH}_3 > \text{PF}_3$
 (B) $\text{BF}_3 > \text{NH}_3 > \text{PF}_3 > \text{I}_3^-$
 (C) $\text{I}_3^- > \text{NH}_3 > \text{PF}_3 > \text{BF}_3$
 (D) $\text{BF}_3 > \text{I}_3^- > \text{PF}_3 > \text{NH}_3$

538. Identify the pair in which the geometry of the species is T-shape and square pyramidal respectively:

- (A) IO_3^- and IO_2F_2^-
 (B) XeOF_2 and XeOF_4
 (C) ICl_2^- and ICl_5
 (D) ClF_3 and IO_4^-

539. Number of compounds with one lone pair of electrons on central atom amongst following is ____.

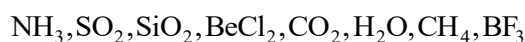


540. Match List-I with List-II

Column-I (Molecule)		Column-II (Shape)	
(A)	BrF_5	(I)	T-shape
(B)	H_2O	(II)	See saw
(C)	ClF_3	(III)	Bent
(D)	SF_4	(IV)	Square pyramidal

- (A) $\text{A} \rightarrow (\text{I}), \text{B} \rightarrow (\text{II}), \text{C} \rightarrow (\text{IV}), \text{D} \rightarrow (\text{III})$
 (B) $\text{A} \rightarrow (\text{II}), \text{B} \rightarrow (\text{I}), \text{C} \rightarrow (\text{III}), \text{D} \rightarrow (\text{IV})$
 (C) $\text{A} \rightarrow (\text{III}), \text{B} \rightarrow (\text{IV}), \text{C} \rightarrow (\text{I}), \text{D} \rightarrow (\text{II})$
 (D) $\text{A} \rightarrow (\text{IV}), \text{B} \rightarrow (\text{III}), \text{C} \rightarrow (\text{I}), \text{D} \rightarrow (\text{II})$

541. The number of species from the following in which the central atom uses sp^3 hybrid orbitals in its bonding is ____.



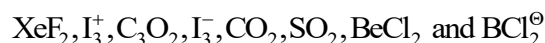
542. For OF_2 molecule consider the following

- (a) Number of lone pairs on oxygen is 2
 (b) FOF angle is less than 104.5°
 (c) Oxidation state of O is -2
 (d) Molecule is bent 'V' shaped
 (e) Molecular geometry is linear

Correct options are:

- (A) c, d, e only (B) b, e, a only
 (C) a, c, d only (D) a, b, d only

543. Amongst the following, the number of species having the linear shape is ____.



544. Match List-I with List-II

Column-I (Molecule/ions)		Column-II (No. of lone pairs of e- on central atom)	
(A)	IF_7	(I)	Three
(B)	ICl_4^-	(II)	One
(C)	XeF_6	(III)	Two
(D)	XeF_2	(IV)	Zero

Choose the correct answer from the options given below:

- (A) $\text{A} \rightarrow (\text{II}), \text{B} \rightarrow (\text{III}), \text{C} \rightarrow (\text{IV}), \text{D} \rightarrow (\text{I})$
 (B) $\text{A} \rightarrow (\text{IV}), \text{B} \rightarrow (\text{III}), \text{C} \rightarrow (\text{II}), \text{D} \rightarrow (\text{I})$
 (C) $\text{A} \rightarrow (\text{II}), \text{B} \rightarrow (\text{I}), \text{C} \rightarrow (\text{IV}), \text{D} \rightarrow (\text{III})$
 (D) $\text{A} \rightarrow (\text{IV}), \text{B} \rightarrow (\text{I}), \text{C} \rightarrow (\text{II}), \text{D} \rightarrow (\text{III})$

545. Match List-I with List-II

Column-I (Molecule)		Column-II (Shape)	
(A)	XeF_4	(I)	See saw
(B)	SF_4	(II)	Square planar
(C)	NH_4^+	(III)	Bent T-shape
(D)	BrF_3	(IV)	Tetrahedral

Choose the correct answer from the options given below:

- (A) $\text{A} \rightarrow (\text{IV}), \text{B} \rightarrow (\text{III}), \text{C} \rightarrow (\text{II}), \text{D} \rightarrow (\text{I})$
 (B) $\text{A} \rightarrow (\text{II}), \text{B} \rightarrow (\text{I}), \text{C} \rightarrow (\text{III}), \text{D} \rightarrow (\text{IV})$
 (C) $\text{A} \rightarrow (\text{IV}), \text{B} \rightarrow (\text{I}), \text{C} \rightarrow (\text{II}), \text{D} \rightarrow (\text{III})$
 (D) $\text{A} \rightarrow (\text{II}), \text{B} \rightarrow (\text{I}), \text{C} \rightarrow (\text{IV}), \text{D} \rightarrow (\text{III})$

546. Number of lone pair (s) of electrons on central atom and the shape of BrF_3 molecule respectively, are:

- (A) 0, triangular planar
(B) 1, pyramidal
(C) 2, bent T-shape
(D) 1, bent T-shape

547. There are four species CO_2 , $\text{N}_3^\ominus$, $\text{NO}_2^\oplus$ and $\text{I}_3^\ominus$. Which of the following statement is correct about these species.

- (A) All are linear and having sp hybridization of central atoms
(B) All are linear but only CO_2 , $\text{N}_3^\ominus$ and $\text{I}_3^\ominus$ have sp hybridization on their central atom
(C) All are linear but only CO_2 , $\text{N}_3^\ominus$ and $\text{NO}_2^\oplus$ have sp hybridization on their central atom
(D) CO_2 , $\text{N}_3^\ominus$ and $\text{NO}_2^\oplus$ are linear but $\text{I}_3^\ominus$ is not

548. Which molecule is T or arrow ($\rightarrow$) shaped.

- (A) BeF_2 (B) BCl_3
(C) NH_3 (D) ClF_3

Dipole Moment & Hydrogen Bonding

549. The compound of Xenon with zero dipole moment is:

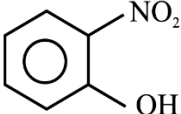
- (A) XeO_3 (B) XeO_2
(C) XeF_4 (D) XeOF_4

550. Assertion (A): PH_3 has lower boiling point than NH_3

Reason (R): In liquid state NH_3 molecules are associated through vander waal's forces, but PH_3 molecules are associated through hydrogen bonding. In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
(B) (A) is not correct but (R) is correct
(C) Both (A) and (R) are correct but (R) is the correct explanation of (A)
(D) (A) is correct but (R) is not correct

551. Select the compound from the following that will show intramolecular hydrogen bonding.

- (A) H_2O (B) NH_3
(C) $\text{C}_2\text{H}_5\text{OH}$ (D) 

552. Choose the polar molecule from the following:

- (A) CCl_4 (B) CO_2
(C) $\text{CH}_2=\text{CH}_2$ (D) CHCl_3

553. The number of non-polar molecules from the following is ____.

$\text{HF}, \text{H}_2\text{O}, \text{SO}_2, \text{H}_2, \text{CO}_2, \text{CH}_4, \text{NH}_3, \text{HCl}, \text{CHCl}_3, \text{BF}_3$

554. Given below are two statements:

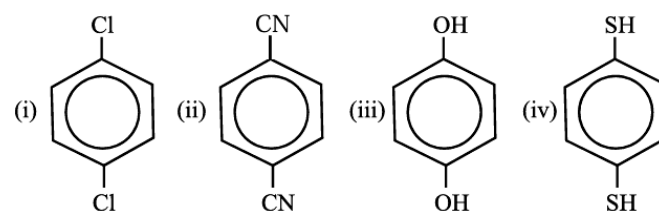
Statement-I: Since fluorine is more electronegative than nitrogen, the net dipole moment of NF_3 is greater than NH_3

Statement-II: In NH_3 , the orbital dipole due to lone pair and the dipole moment of NH bonds are in opposite direction, but in NF_3 the orbital dipole due to lone pair and dipole moment of N-F bonds are in same direction.

In the light of the above statements. Choose the most appropriate from the options given below.

- (A) Statement I is true but Statement II is false
(B) Both Statement I and Statement II are false
(C) Both Statement I and Statement II are true
(D) Statement I is false but Statement II is true

555. For which of the following molecule significant $\mu \neq 0$?



- (A) Only (i)
(B) (i) and (ii)
(C) Only (iii)
(D) (iii) and (iv)

- 556.** If one assumes linear structure instead of bent structure for water, then which one of the following properties cannot be explained?
- (A) The formation of intermolecular hydrogen bond in water.
 (B) The high boiling point of water
 (C) Solubility of polar compounds in water
 (D) Ability of water to form coordinate covalent bond

- 557.** Out of the following compounds, the total number of non polar compounds are:
- (A) ClF (B) Al_2Cl_6
 (C) S_2F_2 (D) SF_2
 (5) SF_4 (6) PCl_2F_3
 (7) PCl_3F_2 (8) ClF_3
 (9) XeF_4 (10) I_2Cl_6
 (11) BF_3

Bent's rule & Drago's rule

- 558.** The decreasing values of bond angles from NH_3 (106°) to SbH_3 (101°) down group-15 of the periodic table is due to
- (A) Decreasing lp-lp repulsion
 (B) Decreasing electronegativity
 (C) Increasing bp-bp repulsion
 (D) Decreasing p-orbital character in sp^3

- 559.** Which statement is correct about bond angle of NCl_3 , NF_3 and NH_3
- I. Bond angle of $\text{NCl}_3 > \text{NF}_3$.
 II. Bond angle of $\text{NCl}_3 < \text{NF}_3$
 III. Bond angle of $\text{NH}_3 > \text{NF}_3$
 IV. Bond angle of $\text{NH}_3 < \text{NF}_3$
- (A) I, II (B) I, III
 (C) I, II, III (D) I, II, IV

- 560.** Decreasing order of bond angle of (I) NO_2 , (II) $\text{NO}_2^\oplus$ and (III) $\text{NO}_3^\ominus$ is
- (A) $\text{I} > \text{II} > \text{III}$ (B) $\text{II} > \text{I} > \text{III}$
 (C) $\text{III} > \text{II} > \text{I}$ (D) $\text{III} > \text{I} > \text{II}$

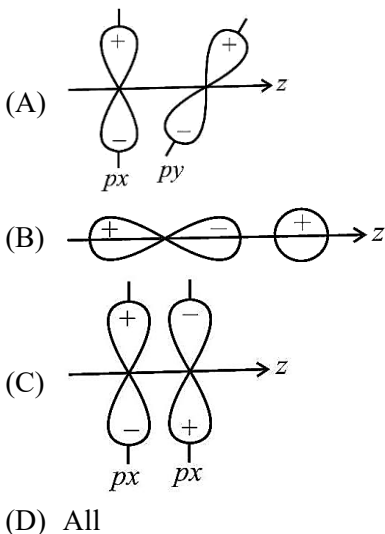
Fajan's rule & Polarisability

- 561.** Which of the following is the most ionic?
- (A) P_4O_{10}
 (B) MnO
 (C) CrO_3
 (D) Mn_2O_7
- 562.** Arrange the bonds in order of increasing ionic character in the molecules. LiF , K_2O , N_2 , SO_2 and ClF_3
- (A) $\text{ClF}_3 < \text{N}_2 < \text{SO}_2 < \text{K}_2\text{O} < \text{LiF}$
 (B) $\text{LiF} < \text{K}_2\text{O} < \text{ClF}_3 < \text{SO}_2 < \text{N}_2$
 (C) $\text{N}_2 < \text{SO}_2 < \text{ClF}_3 < \text{K}_2\text{O} < \text{LiF}$
 (D) $\text{N}_2 < \text{ClF}_3 < \text{SO}_2 < \text{K}_2\text{O} < \text{LiF}$
- 563.** Which of the following is least ionic?
- (A) BaCl_2
 (B) AgCl
 (C) KCl
 (D) CoCl_2

Linear combination of atomic orbitals

- 564.** Given below are two statements:
- Statement-I:** A π bonding MO has lower electron density above and below the inter-nuclear axis.
Statement-II: The π^* antibonding MO has a node between the nuclei.
- In the light of the above statements. Choose the most appropriate from the options given below.
- (A) Both Statement I and Statement II are false
 (B) Both Statement I and Statement II are true
 (C) Statement I is false but Statement II is true
 (D) Statement I is true but Statement II is false
- 565.** The total number of anti bonding molecular orbitals, formed from 2s and 2p atomic orbitals in a diatomic molecule is ____.
- 566.** The total number of molecular orbitals formed from 2s and 2p atomic orbitals of a diatomic molecule

567. Which of the following is a zero overlap which leads to non-bonding?



568. Which of the following overlap is correct (assuming Z-axis as internuclear axis)?

- (I) $2p_x + 2p_x \rightarrow \sigma$ -bond
 (II) $2s + 2p_y \rightarrow \pi$ -bond
 (III) $3d_{xy} + 3d_{xy} \rightarrow \pi$ -bond
 (IV) $3d_{yz} + 3d_{yz} \rightarrow \pi$ -bond
 (V) $2p_y + 2p_y \rightarrow \pi$ -bond
 (VI) $2p_z + 2p_z \rightarrow \sigma$ -bond

- (A) I, II, III (B) IV, V, VI
 (C) I, III, V (D) II, IV, VI

569. The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals

- (a) Have the same energy
 (b) Have the minimum overlap
 (c) Have same symmetry about the molecular axis
 (d) Have different symmetry about the molecular axis

Choose the most appropriate from the options given below:

- (A) a, b, c only (B) a and c only
 (C) b, c, d only (D) b and d only

570. Which of the following MO's has zero nodal planes?

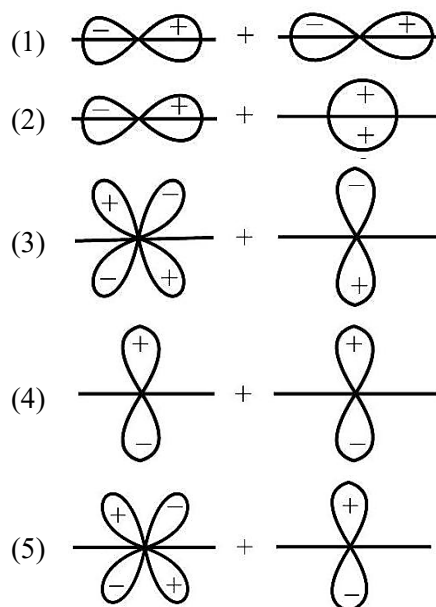
- (A) $\sigma^* 1s$ (B) $\sigma 1s$
 (C) $\pi 2p_x$ (D) $\pi^* 2p_x$

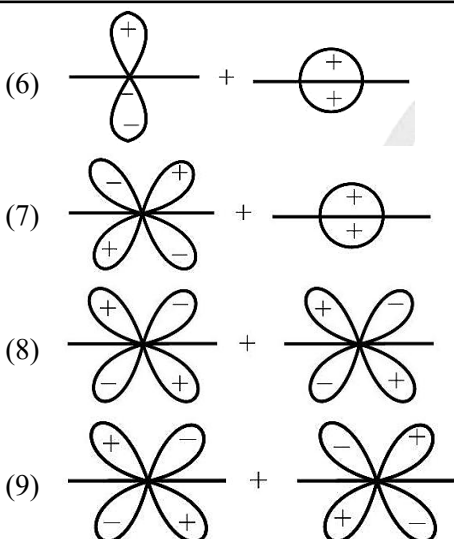
571. Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists.

List-I		List-II	
(P)		(1)	p - d π antibonding
(Q)		(2)	d - d σ bonding
(R)		(3)	p - d π bonding
(S)		(4)	d - d σ antibonding

- (A) P $\rightarrow$ (2), Q $\rightarrow$ (1), R $\rightarrow$ (3), S $\rightarrow$ (4)
 (B) P $\rightarrow$ (4), Q $\rightarrow$ (3), R $\rightarrow$ (1), S $\rightarrow$ (2)
 (C) P $\rightarrow$ (2), Q $\rightarrow$ (3), R $\rightarrow$ (1), S $\rightarrow$ (4)
 (D) P $\rightarrow$ (4), Q $\rightarrow$ (1), R $\rightarrow$ (3), S $\rightarrow$ (2)

572. Some arrangement of atomic orbitals are given below:

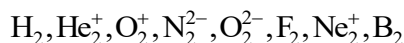




If A, B and C are number of arrangements which give bonding molecular orbitals, anti-bonding molecular orbitals and non bonding molecular orbitals respectively. Then the value of (A+B+C) is:

Molecular Orbital Theory

573. The number of species from the following which are paramagnetic and with bond order equal to one is ____.



574. What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : $\text{N}_2 : \text{N}_2^+; \text{O}_2; \text{O}_2^+$?

- (A) 0,1,2,1 (B) 2,1,2,1
(C) 0,1,0,1 (D) 2,1,0,1

575. According to MO theory the bond orders for O_2^{2-} , CO and NO^+ respectively, are

- (A) 1,3 and 3 (B) 1,3 and 2
(C) 1,2 and 3 (D) 2,3 and 3

576. The correct order of bond orders of C_2^{2-} , N_2^{2-} and O_2^{2-} is, respectively.

- (A) $\text{C}_2^{2-} < \text{N}_2^{2-} < \text{O}_2^{2-}$
(B) $\text{O}_2^{2-} < \text{N}_2^{2-} < \text{C}_2^{2-}$
(C) $\text{C}_2^{2-} < \text{O}_2^{2-} < \text{N}_2^{2-}$
(D) $\text{N}_2^{2-} < \text{C}_2^{2-} < \text{O}_2^{2-}$

577. Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO theory, by removal of an electron?

- (a) NO (b) N_2
(c) O_2 (d) C_2
(e) B_2

Choose the most appropriate answer from the options given below:

- (A) (a), (b), (c) only (B) (b), (c), (e) only
(C) (a), (c) only (D) (d) only

578. Out of the following species, the total number of species which have symmetrical electron distribution in their HOMO (Highest Occupied Molecular Orbital) and also paramagnetic are:

- (1) C_2^{2-} (2) B_2
(3) O_2 (4) N_2^{2-}
(5) C_2 (6) $\text{N}_2^{\oplus}$
(7) O_2^{2-}

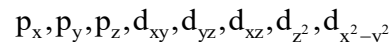
579. Which of the following statement is not correct from the view point of molecular orbital theory?

- (A) Be_2 is not a stable molecule
(B) He_2 is not stable but He_2^+ is expected to exist
(C) Bond strength of N_2 is maximum amongst the homonuclear diatomic molecules belonging to the second period
(D) The order of energies of molecular orbitals in N_2 molecule is

$$\sigma_{2s} < \sigma_{2s}^* < \sigma_{2pz} < (\pi_{2px} = \pi_{2py}) < (\pi_{2px}^* = \pi_{2py}^*) < \sigma_{2pz}^*$$

Shapes of Molecular Orbital

580. The number of given orbitals which have electron density along the axis is ____.



Miscellaneous

581. The number of correct options is

- (i) $\text{I}^{\ominus} > \text{Br}^{\ominus} > \text{Cl}^{\ominus} > \text{F}^{\ominus}$ (polarisability)
(ii) $\text{Li}^{\oplus} > \text{Na}^{\oplus} > \text{K}^{\oplus} > \text{Rb}^{\oplus}$ (polarization power)
(iii) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$ (Order of b.pt.)
(iv) $\text{H}_2^{\ominus} < \text{H}_2^{\oplus}$ (order of stability)

- 582.** Which of the following statements is incorrect?
 (A) NH_3 is more basic than PH_3
 (B) NH_3 has a higher boiling point than that of HF
 (C) N_2 is less reactive than P_4
 (D) The dipole moment of NH_3 is less than that of SO_2

NEW QUESTIONS

- 583.** In which one of the following compounds does the central atom obey the octet rule?
 (A) XeF_4 (B) XeOF_2
 (C) SCl_2 (D) AlCl_3
- 584.** The attractive force which holds various constituents together in different chemical species is called
 (A) valence bond
 (B) chemical bond
 (C) atomic bond
 (D) electrovalent bond
- 585.** The octet rule is observed in
 (A) PCl_5 (B) CO_2
 (C) BCl_3 (D) SF_6
- 586.** Which one of the following contains ionic, covalent and co-ordinate bonds?
 (A) NaOH (B) NaCl
 (C) HCl (D) NaNC
- 587.** Which of the following is correct for the process of formation of a chemical bond?
 (A) Electron-electron repulsion becomes more than the nucleus-electron attraction
 (B) Electron-electron repulsion becomes more than the nucleus-electron attraction
 (C) Energy increases.
 (D) Energy decreases.
- 588.** Example of super octet molecule is
 (A) SF_6 (B) PCl_5
 (C) IF_7 (D) All of these

- 589.** The number of electrons involved in the bond formation in N_2 molecule is
 (A) 2 (B) 4
 (C) 10 (D) 6
- 590.** The octet rule is not obeyed in
 (A) CO_2 (B) BCl_3
 (C) PCl_5 (D) (B) and (C) both
- 591.** NH_3 and BF_3 combine readily because of the formation of
 (A) a covalent bond (B) a hydrogen bond
 (C) a coordinate bond (D) an ionic bond
- 592.** Which of the following species are hypervalent?
 1. ClO_4^- 2. BF_3
 3. SO_4^{2-} 4. CO_3^{2-}
 (A) 1,2,3 (B) 1,3
 (C) 3,4 (D) 1,2
- 593.** If the atomic number of element X is 7 the Lewis diagram for the element is
 (A) $\text{X} \cdot$ (B) $\cdot \ddot{\text{X}} \cdot$
 (C) $\cdot \ddot{\text{X}} :$ (D) $:\ddot{\text{X}}:$
- 594.** According to Fajans rule covalent bond is favoured by
 (A) Small cation and large anion
 (B) Small cation and small anion
 (C) Large cation and large anion
 (D) Large cation and small anion
- 595.** Polarisation is the distortion of the shape of anion by an adjacently placed cation. Which of the following statement is correct?
 (A) Maximum polarization is brought about by cation of high charge
 (B) Maximum polarization is brought about by cation of Low charge
 (C) A large cation is likely to bring about a large degree of polarization
 (D) A small anion is likely undergo a large degree of polarization

- 596.** Polarization power of a cation increases when
 (A) Charge on the cation increases
 (B) Size of the cation increases
 (C) Charge on the cation decreases
 (D) Has no relation to its size or charge
- 597.** In a covalent bond formation,
 (A) transfer of electrons takes place
 (B) equal sharing of electrons between two atoms takes place
 (C) electrons are shared by one atom only
 (D) electrons are donated by one atom and shared by both atoms.
- 598.** Which of the following shows the Lewis dot formula for CO_2 ?
 (A) $\text{:}\ddot{\text{O}}\text{:C}::\ddot{\text{O}}\text{:}$ (B) $\text{:}\ddot{\text{O}}\text{:}\ddot{\text{C}}::\ddot{\text{O}}\text{:}$
 (C) $\text{:}\ddot{\text{O}}\text{:C}:\ddot{\text{O}}\text{:}$ (D) $\text{:}\ddot{\text{O}}\text{:C}:\ddot{\text{O}}\text{:}$
- 599.** Among the following the exceptions of the octet rule is:
 (A) the incomplete octet of central atom
 (B) an odd number of electrons on central atom
 (C) expanded octet of the central atom
 (D) all of these
- 600.** In which of the following species the bond is nondirectional?
 (A) NCl_3 (B) RbCl
 (C) BeCl_2 (D) BCl_3
- 601.** Sodium chloride has a crystalline structure made up of Na_+ and Cl_- ions. Why does NaCl not conduct electricity in solid state?
 (A) The ions of NaCl become mobile only in molten state and are not free to move in solid state
 (B) The crystalline structure does not have ions
 (C) When a bond is formed between ions they lose their charges
 (D) None of these
- 602.** Most ionic compounds have:
 (a) High melting point and low boiling point
 (b) High melting point and non-directional bonds
 (c) High solubilities in polar solvents and low solubilities in non-polar solvents
 (d) Three dimensional network structures and are good conductors of electricity in the molten state
 (A) None of the above (B) b, c, d
 (C) a, b, c (D) All of the above
- 603.** In an ionic compound A^+X^- the degree of covalent bonding is greatest when:
 (A) A^+ and X^- ion are small
 (B) A^+ and X^- ion are large
 (C) A^+ and X^- ion are approximately of the same size
 (D) A^+ is small and X^- is large
- 604.** Which one of the following has the highest lattice energy?
 (A) LiCl (B) CaCl_2
 (C) LiF (D) KCl
- 605.** Lattice energy of a solid increases if:
 (A) The radii of ions are small
 (B) The ions are neutral
 (C) Charges on the ions are small
 (D) Madelung constant decreases
- 606.** The decreasing order of ionic nature of the following compound is:
 (A) $\text{LiI} > \text{NaBr} > \text{KCl} > \text{CsF}$
 (B) $\text{LiI} > \text{KCl} > \text{NaBr} > \text{CsF}$
 (C) $\text{CsF} > \text{NaBr} > \text{KCl} > \text{LiI}$
 (D) $\text{CsF} > \text{KCl} > \text{NaBr} > \text{LiI}$
- 607.** The most covalent halide of s-Block is:
 (A) CsF (B) CsBr
 (C) BeI_2 (D) BeF_2
- 608.** The ionic compounds will be sparingly soluble if:
 (A) Hydration energy = lattice energy
 (B) Hydration energy > lattice energy
 (C) Hydration energy < lattice energy
 (D) Hydration energy = 0

- 609.** Predict the correct order of repulsion among the following:
 (A) bond pair - bond pair > lone pair - bond pair
 lone pair - lone pair
 (B) lone pair - bond pair > bond pair - bond pair
 lone pair - lone pair
 (C) lone pair - lone pair > lone pair - bond pair
 > bond pair - bond pair
 (D) lone pair - lone pair > bond pair - bond pair
 lone pair - bond pair
- 610.** Which of the following molecules has more than one lone pair?
 (A) SO_2 (B) XeF_2
 (C) SiF_4 (D) CH_4
- 611.** In the structure of ClF_3 , the number of lone pairs of electrons on central atom 'Cl' is:
 (A) one (B) two
 (C) four (D) three
- 612.** The central atom of which one of the following molecules, has 3 bond pairs and 2 lone pairs of electrons in its valence shell?
 (A) Boron trifluoride
 (B) Phosphorous trichloride
 (C) Chlorine trifluoride
 (D) Xenon trioxide
- 613.** Number of b.p. and l.p. in NO^- is:
 (A) 3 b.p. +1 l.p. (B) 4 b.p. +0 l.p.
 (C) 2 b.p. +2 l.p. (D) 1 b.p. +1 l.p.
- 614.** SF_4 can be classified as
 (A) AB_4 type (B) AB_4E type
 (C) AB_4E_2 type (D) None of these
- 615.** Pentagonal bipyramidal structure contains bond angles approximately
 (A) $120^\circ, 90^\circ, 180^\circ$ (B) $120^\circ, 72^\circ, 180^\circ$
 (C) $72^\circ, 90^\circ, 120^\circ$ (D) $72^\circ, 90^\circ, 180^\circ$
- 616.** In which of the following molecules, the central atom has two lone pairs of electrons?
 (A) SF_4 (B) BrF_5
 (C) SO_2 (D) XeF_4
- 617.** Among the following molecules: $\text{SO}_2, \text{SF}_4, \text{ClF}_3$, BrF_5 and XeF_4 which of the following shapes does not describe any of the molecules mentioned?
 (A) Bent
 (B) Trigonal bipyramidal
 (C) See-saw
 (D) T-shape
- 618.** In which of the following molecule/ion, all the bonds are not equal?
 (A) XeF_4 (B) BF_4^-
 (C) C_2H_4 (D) SiF_4
- 619.** Incorrect matching amongst the following is:
 (i) Linear - $22\text{HO}, \text{SO}$
 (ii) V-shaped - $22 : \text{CH}, \text{SnCl}$
 (iii) See Saw - $44 \text{SF}, \text{TeCl}$
 (iv) T-shaped- $33 \text{ICl}, \text{ClF}$
 (A) only (i) (B) both (ii and (ii)
 (C) only (iii) (D) both (iii) and (iv)
- 620.** According to VSEPR theory molecule having Tshape is:
 (A) 3 ClF (B) 3 NH
 (C) 3 BF (D) 2 HO
- 621.** Which of the following has a regular geometry?
 (A) CHCl_3 (B) PCl_3
 (C) XeF_6 (D) SF_4
- 622. Assertion:** ClF_3 has T-shape structure.
Reason: It has two lone pairs arranged at 180° angle.
 (A) If both assertion and reason are true and reason is the correct explanation of assertion.
 (B) If both assertion and reason are true but reason is not the correct explanation of assertion.
 (C) If assertion is true but reason is false.
 (D) If both assertion and reason are false.

- 623.** Which of the following molecules is linear in shape?
 (A) XeF_6 (B) XeO_3
 (C) XeF_4 (D) XeF_2
- 624.** The shape of is: 2 XeF
 (A) Linear (B) Bent
 (C) T-shaped (D) None of these
- 625.** Which of the following is planar?
 (A) XeO_4 (B) $\text{XeO}_2 \text{F}_2$
 (C) $\text{XeO}_3 \text{F}_2$ (D) XeF_4
- 626.** Which one of the following molecules are formed by p-p overlapping?
 (A) Cl_2 (B) HCl
 (C) H_2O (D) NH_3
- 627.** The bond in the formation of fluorine molecule will be:
 (A) Due to s-s overlapping
 (B) Due to s-p overlapping
 (C) Due to p-p overlapping
 (D) Due to hybridization
- 628.** Which type of overlapping results the formation of a π -bond
 (A) Axial overlapping of s-s orbitals
 (B) Lateral overlapping of p-p orbitals
 (C) Axial overlapping of p-p orbitals
 (D) Axial overlapping of s-p orbitals
- 629.** In a sigma bond
 (A) Sidewise as well as end to end overlap of orbitals take place
 (B) Sidewise overlap of orbitals takes place
 (C) End to end overlap of orbitals takes place
 (D) None of the above
- 630.** Which is correct option for overlapping strength
 (A) $2s-2s < 2s-2p < 2p-2p$
 (B) $1s-1s < 1s-2p < 1p-2p$
 (C) $2s-2s < 1s-2p < 1s-1s$
 (D) Both (A) and (C)
- 631.** Which overlapping consist of maximum overlapped area between nucleus:
 (A) $2s-2s$ (B) $2p-2p$
 (C) $2s-2p$ (D) Both (A) and (C)
- 632.** The species, having bond angles of 120° is
 (A) ClF_3 (B) NCl_3
 (C) BCl_3 (D) PH_3
- 633.** Consider the molecules CH_4 , NH_3 and H_2O . Which of the given statements is false?
 (A) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is smaller than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
 (B) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 is larger than the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 .
 (C) The $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 , the $\text{H}-\text{N}-\text{H}$ bond angle in NH_3 , and the $\text{H}-\text{O}-\text{H}$ bond angle in H_2O are all greater than 90° .
 (D) The $\text{H}-\text{O}-\text{H}$ bond angle in H_2O is larger than the $\text{H}-\text{C}-\text{H}$ bond angle in CH_4 .
- 634.** The molecule having smallest bond angle is:
 (A) AsCl_3 (B) SbCl_3
 (C) PCl_3 (D) NCl_3
- 635.** In which molecule are all atoms coplanar
 (A) CH_4 (B) BF_3
 (C) PF_3 (D) NH_3
- 636.** Which has the least bond angle:
 (A) NH_3 (B) BeF_2
 (C) H_2O (D) CH_4
- 637.** The correct order of bond angles is:
 (A) $\text{NO}_2^- > \text{NO}_2^+ > \text{NO}_2$
 (B) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$
 (C) $\text{NO}_2 > \text{NO}_2^+ > \text{NO}_2^-$
 (D) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$

- 638.** The correct order of H–M–H bonds angle is:
 (A) $\text{NH}_3 < \text{PH}_3 < \text{SbH}_3 < \text{BiH}_3$
 (B) $\text{AsH}_3 < \text{SbH}_3 < \text{PH}_3 < \text{NH}_3$
 (C) $\text{NH}_3 < \text{PH}_3 < \text{BiH}_3 < \text{SbH}_3$
 (D) $\text{BiH}_3 < \text{SbH}_3 < \text{AsH}_3 < \text{PH}_3$
- 639.** The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is
 (A) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$
 (B) $\text{H}_2\text{S} < \text{NH}_3 < \text{BF}_3 < \text{SiH}_4$
 (C) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$
 (D) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
- 640.** Which of the following is the correct order of dipole moment?
 (A) $\text{NH}_3 < \text{BF}_3 < \text{NF}_3 < \text{H}_2\text{O}$
 (B) $\text{BF}_3 < \text{NF}_3 < \text{NH}_3 < \text{H}_2\text{O}$
 (C) $\text{BF}_3 < \text{NH}_3 < \text{NF}_3 < \text{H}_2\text{O}$
 (D) $\text{H}_2\text{O} < \text{NF}_3 < \text{NH}_3 < \text{BF}_3$
- 641.** Among the following molecules, which has the zero-dipole moment?
 (A) BF_3 (B) H_2O
 (C) NF_3 (D) ClO_2
- 642.** Zero dipole moment is possessed by:
 (A) PCl_3 (B) BF_3
 (C) ClF_3 (D) NH_3
- 643.** Which one of the following is a non-polar molecule?
 (A) CCl_4 (B) CHCl_3
 (C) CH_2Cl_2 (D) CH_3Cl
- 644.** Identify the non-polar molecule in the set of compounds given:
 (A) H_2 (B) HCl
 (C) HF, HBr (D) HBr
- 645.** The bond angle and dipole moment of water respectively, are:
 (A) $109.5^\circ, 1.84\text{D}$ (B) $107.5^\circ, 1.56\text{D}$
 (C) $104.5^\circ, 1.84\text{D}$ (D) $102.5^\circ, 1.56\text{D}$
- 646.** Which of the following has dipole moment?
 (A) CO_2
 (B) p-dichlorobenzene
 (C) NH_3
 (D) CH_4
- 647.** The decreasing order of dipole moment of molecules is
 (A) $\text{NF}_3 > \text{NH}_3 > \text{H}_2\text{O}$
 (B) $\text{NH}_3 > \text{NF}_3 > \text{H}_2\text{O}$
 (C) $\text{H}_2\text{O} > \text{NH}_3 > \text{NF}_3$
 (D) $\text{H}_2\text{O} > \text{NF}_3 > \text{NH}_3$
- 648.** H_2O has a net dipole moment while BeF_2 has zero dipole moment because:
 (A) F is more electronegativity than oxygen
 (B) Be is more electronegativity than oxygen
 (C) H_2O molecule is linear and BeF_2 is bent
 (D) BeF_2 molecule is linear and H_2O is bent
- 649.** Resonance is not shown by:
 (A) C_6H_6 (B) NO_2
 (C) CO_3^{2-} (D) H_2O
- 650.** BeF_2 has zero dipole moment whereas H_2O has dipole moment because:
 (A) Water is linear
 (B) H_2O is bent
 (C) F is more electronegative than O
 (D) Hydrogen bonding is present in H_2O
- 651.** Which of the following species are polar:
 (1) C_6H_6 (2) XeF_2
 (3) SO_2 (4) SF_4
 (5) SF_6
 Correct answer is:
 (A) (1) & (4) (B) (1), (2) & (5)
 (C) (1) & (5) (D) (3) & (4)
- 652.** Dipole moment is shown by:
 (A) 1, 4-Dichlorobenzene
 (B) Cis 1, 2-dichloroethene
 (C) Trans -1, 2-dichloro ethene
 (D) Benzene

- 653.** What conclusion can be drawn from the fact that BF_3 has no dipole moment but PF_3 does:
 (A) BF_3 is not symmetrical but PF_3 is symmetrical
 (B) BF_3 molecule must be linear
 (C) Atomic radius of P is larger than that of B
 (D) BF_3 molecule must be planar triangular
- 654.** A molecule MX_4 has a square planar shape. The number of non-bonding pair of electrons is
 (A) 2 (B) 1
 (C) 3 (D) 0
- 655.** The percentage of s-character in the hybridized orbitals of B in BF_3 is:
 (A) 25 (B) 50
 (C) 75 (D) 33.3
- 656.** In which of the following pairs both the species have sp^3 hybridization?
 (A) $\text{SiF}_4, \text{BeH}_2$ (B) $\text{NF}_3, \text{H}_2\text{O}$
 (C) NF_3, BF_3 (D) $\text{H}_2\text{S}, \text{BF}_3$
- 657.** Which of the following statements is correct?
 (A) sp^3 hybrid orbitals have equal s and p character
 (B) The bond angle decreases with the decrease of s character of hybridized orbital
 (C) Resonance decreases the stability of a molecule
 (D) Resonance is due to delocalization of sigma electrons
- 658.** The percentage s-character of the hybrid orbitals in methane, ethane and ethyne are respectively:
 (A) 25, 33, 50 (B) 25, 50, 75
 (C) 50, 75, 100 (D) 10, 20, 40
- 659.** The percentage of p-character of the hybrid orbitals in graphite and diamond are respectively:
 (A) 33 and 25 (B) 50 and 75
 (C) 67 and 75 (D) 33 and 75
- 660.** Pick out the incorrect statement from the following:
 (A) sp hybrid orbitals are equivalent and are at an angle of 180° with each other.
 (B) sp^2 hybrid orbitals are equivalent and bond angle between any two of them is 120°
 (C) $\text{sp}^3 \text{d}^2$ hybrid orbitals are equivalent and are oriented towards corners of a regular octahedron.
 (D) $\text{sp}^3 \text{d}^3$ hybrid orbitals are not equivalent
- 661.** The hybridization of carbon in molecule CO_2 is:
 (A) sp (B) sp^2
 (C) sp^3 (D) $\text{sp}^3 \text{d}$
- 662.** What is the type of hybridisation found in methane?
 (A) sp^3 (B) sp^2
 (C) sp^1 (D) None of these
- 663.** The hybridization at carbons involved in C^*-C^* bond in $\text{HC}\equiv\text{C}^*-\text{CH}^*=\text{CH}_2$ is :
 (A) sp^3-sp^2 (B) sp^3-sp^3
 (C) $\text{sp}-\text{sp}^2$ (D) $\text{sp}-\text{sp}^3$
- 664.** The hybridisation and shape of ClO_3^- is :
 (A) sp^3 and tetrahedral
 (B) sp^3 and triangular pyramidal
 (C) sp^3 and triangular planar
 (D) $\text{sp}^3 \text{d}$ and trigonal bipyramidal
- 665.** The state of hybridization of Xe in XeF_4 is:
 (A) sp^2 (B) sp^3
 (C) $\text{sp}^3 \text{d}$ (D) $\text{sp}^3 \text{d}^2$
- 666.** sp^2 -hybridization is shown by:
 (A) BeCl_2 (B) BF_3
 (C) NH_3 (D) XeF_2

- 667.** Carbon atoms in $C_2(CN)_4$ are:
 (A) sp-hybridized
 (B) sp^2 -hybridized
 (C) sp and sp^2 -hybridized
 (D) sp, sp^2 and sp^3 - hybridized
- 668.** Among the following which species has same number of σ and π -bond?
 (A) C_7H_8 (B) $C_2(CN)_4$
 (C) C_2H_4 (D) $HC \equiv CH$
- 669.** Which one of the following is the correct set with respect to molecule, hybridization and shape?
 (A) $BeCl_2, sp^2$, linear
 (B) $BeCl_2, sp^2$, triangular planar
 (C) BCl_3, sp^2 , triangular planar
 (D) BCl_3, sp^3 , tetrahedral
- 670.** The hybridization of central iodine atom in IF_5, I_3^- and I_3^+ are respectively:
 (A) sp^3d^2, sp^3d, sp^3 (B) sp^3d, sp^3d, sp^3
 (C) sp^3d^2, sp^3d^2, sp^3 (D) sp^3d, sp^3d^2, sp^3
- 671.** Position of fluorine in $PCl_3 F_2$
 (A) Both at equatorial position
 (B) Both at axial position
 (C) 1 at equatorial and 1 at axial position
 (D) Cannot be predicted
- 672.** In $sp^3 d$ hybridization with 2 lone pairs, the lone pairs are placed at ____ position and shape of molecule is ____.
 (A) Equatorial, trigonal bipyramidal
 (B) Axial, trigonal bipyramidal
 (C) Equatorial, Bent-T
 (D) Axial, Trigonal planar
- 673.** %s character in equatorial orbital in TBP is:
 (A) 50% (B) 25%
 (C) 33.33% (D) 0%
- 674.** Bond angle in PH_3 is:
 (A) Close to 109.28° (B) Same as NH_3
 (C) Close to 90° (D) Close to 120°
- 675.** Hybridization of PBr_5 in the solid state is:
 (A) sp^3 (B) sp^3d
 (C) sp^2 (D) sp
- 676.** Each carbon in carbon suboxide is
 (A) sp^2 -hybridized
 (B) sp^3 -hybridized
 (C) sp-hybridized
 (D) sp^3 -hybridized but linked with one coordinate bond
- 677.** For BF_3 molecule which of the following is true?
 (A) B-atom is sp^2 hybridised
 (B) There is a $p\pi-p\pi$ back bonding in this molecule
 (C) Observed B-F bond length is found to be less than the expected bond length
 (D) All of these
- 678.** Respective order of strength of back-bonding in boron trihalides is
 (A) $BF_3 < BCl_3 < BBr_3$
 (B) $BF_3 > BCl_3 > BBr_3$
 (C) $BBr_3 > BCl_3 > BF_3$
 (D) $BF_3 < BCl_3 < BBr_3$
- 679.** The bond length in O_2^+, O_2, O_2^- and O_2^{2-} follows the order:
 (A) $O_2^{2-} > O_2^- > O_2 > O_2^+$
 (B) $O_2^+ > O_2 > O_2^- > O_2^{2-}$
 (C) $O_2 > O_2^- > O_2^{2-} > O_2^+$
 (D) $O_2^- > O_2^{2-} > O_2^+ > O_2$

680. Which of the following of pair species has the bond order equal to $1/2$?

- (A) H_2, H_2^+ (B) He_2^+, He
(C) $\text{He}_2, \text{H}_2^+$ (D) $\text{H}_2^+, \text{He}_2^+$

681. If a homonuclear diatomic molecule is oriented along the Z -axis, the molecular orbital formed by linear combination of p_x , orbitals of the two atoms is:

- (A) σ (B) σ^*
(C) π (D) δ

682. The correct statement among the following is

- (A) N_2 has higher bond order than N_2^+ and hence has larger bond length compared to N_2^+
(B) N_2^+ has higher bond order than N_2 and hence has larger bond length compared to N_2
(C) N_2 has higher bond order than N_2^+ and hence has higher dissociation energy compared to N_2^+
(D) N_2 has lower bond order than N_2^+ and hence has lower dissociation energy compared to N_2^+ energy

683. The highest occupied MO in N_2 and O_2^+ respectively are (take x -axis as internuclear axis)

- (A) $\sigma 2p_x, \pi^* 2p_y$
(B) $\pi 2p_y, \pi 2p_z$
(C) $\sigma^* 2p_x, \sigma 2p_x$
(D) $\pi^* 2p_y, \pi^* 2p_z$

684. A molecular orbital of a diatomic molecule changes sign when it is rotated by 180° around the molecular axis. This orbital is

- (A) σ (B) π
(C) δ (D) ϕ

685. The ground state electronic configuration of C_2 using all electrons is

- (A) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \sigma_{2p}^2$
(B) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \sigma_{2p}^2 \sigma_{2p}^{*2}$
(C) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^2 \sigma_{2p}^1 \sigma_{2p}^{*1}$
(D) $\sigma_{1s}^2 \sigma_{1s}^{*2} \sigma_{2s}^2 \sigma_{2s}^{*2} \pi_{2p}^4$

686. Match the following items of column I with the appropriate items in column II.

Column I	Column II
Molecular species	Bond order and magnetic
P. O_2^-	I. 2.5 paramagnetic
Q. O_2	II. 2.0 diamagnetic
R. O_2^+	III. 1.5 diamagnetic
S. O_2^{-2}	IV. 1.0 diamagnetic
	V. 2.0 paramagnetic
	VI. 1.5 paramagnetic
(A) P-III, Q-V, R-IV, S-III	
(B) P-III, Q-II, R-I, S-IV	
(C) P-III, Q-V, R-IV, S-VI	
(D) P-VI, Q-V, R-I, S-IV	

687. The bond order of C_2 molecule is

- (A) 0 (B) 1
(C) 2 (D) 3

688. Which of the following has same bond order as NO^+ has?

- (A) CN^- (B) O_2^-
(C) CN^+ (D) None of them

689. Which of the following statements are correct?

- I. Bond order of NO is 2.5
II. Bond order of NO^+ is 3.0
III. Bond order of O_2 is 1.5
IV. Bond order of CO is 3.0
(A) I, II, III (B) I, II, IV
(C) I, III, IV (D) II, III, IV

- 690.** According to MO theory which of the following statement about the magnetic character and bond O_2^+ ?
- (A) Paramagnetic and bond order $< O_2$
 (B) Paramagnetic and bond order $> O_2$
 (C) Diamagnetic and bond order $< O_2$
 (D) Diamagnetic and bond order $> O_2$
- 691.** The correct order in which the O—O bond length increases in the following is:
- (A) $O_2 < O_3 < H_2O_2$ (B) $H_2O_2 < O_3 < O_2$
 (C) $O_3 < O_2 < H_2O_2$ (D) $O_2 < H_2O_2 < O_3$
- 692.** Which one of the following species is paramagnetic?
- (A) O_2^- (B) CN^-
 (C) CO (D) NO^+
- 693.** Molecular orbital electronic configuration for X_2^{n-} anion is:
- $$KK^*(\sigma 2s)^2(\sigma 2s)^2(\pi 2p_x)^2(\pi 2p_y)^2(\sigma 2p_z)^2(\pi 2p_x)^1$$
- The X_2^{n-} is :
- (A) N_2^- (B) O_2^-
 (C) N_2^{2-} (D) O_2^{2-}
- 694.** Which of the following MO has two nodal planes perpendicular to each other
- (A) σ_{2s} (B) $\pi 2p_x$
 (C) $\pi^* 2p_y$ (D) $\pi^* 2p_z$
- 695.** Which pair among the following has identical bond order?
- (A) CN^-, CO^+ (B) O_2^-, O_2^+
 (C) CN^-, CN^+ (D) NO^+, CN^-

Answer Key**Octet Rule, limitations of octet rule, The Lewis Structures, Multiple Covalent Bonds, Ionic or Electrovalent Bonds, Formal charge**

- | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|
| 1. (C) | 2. (C) | 3. (B) | 4. (B) | 5. (A) | 6. (A) | 7. (D) |
| 8. (B) | 9. (B) | 10. (C) | 11. (D) | 12. (B) | 13. (B) | 14. (D) |
| 15. (C) | 16. (A) | 17. (D) | 18. (C) | 19. (C) | 20. (B) | 21. (D) |
| 22. (A) | 23. (B) | 24. (A) | 25. (B) | 26. (C) | 27. (D) | 28. (C) |
| 29. (D) | 30. (C) | 31. (D) | 32. (D) | 33. (D) | 34. (C) | 35. (C) |
| 36. (B) | 37. (B) | 38. (C) | 39. (C) | 40. (A) | 41. (A) | 42. (A) |
| 43. (B) | 44. (A) | 45. (A) | 46. (D) | 47. (B) | 48. (C) | |

Ionic Bond: Factors which favour the formation of ionic bond, Lattice Enthalpy, Hydration Energy, Solubility of Ionic Salt, Thermal Stability

- | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|
| 49. (D) | 50. (B) | 51. (D) | 52. (D) | 53. (D) | 54. (B) | 55. (D) |
| 56. (B) | 57. (B) | 58. (A) | 59. (D) | 60. (B) | 61. (D) | 62. (A) |
| 63. (C) | 64. (D) | 65. (B) | | | | |

Bond Parameter: Bond Length, Bond Angle, Bond Enthalpy

- | | | | | | | |
|---------|---------|---------------|---------|---------|---------|---------|
| 66. (C) | 67. (C) | 68. (D) | 69. (C) | 70. (A) | 71. (B) | 72. (A) |
| 73. (C) | 74. (D) | 75. (A) | 76. (D) | 77. (B) | 78. (B) | 79. (C) |
| 80. (C) | 81. (B) | 82. (D) | 83. (B) | 84. (C) | 85. (A) | 86. (A) |
| 87. (D) | 88. (B) | 89. (A, C, D) | | 90. (A) | 91. (B) | 92. (A) |
| 93. (D) | 94. (D) | 95. (D) | 96. (D) | 97. (A) | 98. (C) | |

Resonating Structures, Dipole Moment

- | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|
| 99. (C) | 100. (D) | 101. (B) | 102. (A) | 103. (C) | 104. (D) | 105. (D) |
| 106. (A) | 107. (D) | 108. (A) | 109. (B) | 110. (B) | 111. (C) | 112. (C) |
| 113. (B) | 114. (A) | 115. (A) | 116. (C) | 117. (C) | 118. (D) | 119. (C) |
| 120. (C) | 121. (B) | 122. (D) | 123. (B) | 124. (C) | 125. (C) | 126. (B) |
| 127. (A) | 128. (D) | 129. (D) | 130. (A) | 131. (B) | 132. (A) | 133. (C) |
| 134. (B) | 135. (C) | 136. (C) | 137. (D) | 138. (B) | 139. (A) | 140. (B) |
| 141. (C) | 142. (C) | 143. (B) | 144. (C) | 145. (B) | 146. (C) | 147. (A) |
| 148. (B) | 149. (A) | 150. (C) | 151. (C) | 152. (B) | 153. (D) | 154. (B) |
| 155. (A) | 156. (D) | 157. (B) | 158. (D) | 159. (A) | 160. (B) | 161. (D) |
| 162. (A) | | | | | | |

VSEPR Theory

- | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|
| 163. (B) | 164. (B) | 165. (C) | 166. (B) | 167. (B) | 168. (D) | 169. (D) |
| 170. (B) | 171. (C) | 172. (A) | 173. (A) | 174. (A) | 175. (C) | 176. (D) |
| 177. (A) | 178. (D) | 179. (C) | 180. (D) | 181. (A) | 182. (C) | 183. (C) |
| 184. (D) | 185. (C) | 186. (D) | 187. (D) | 188. (D) | 189. (A) | 190. (D) |
| 191. (C) | 192. (C) | 193. (B) | 194. (C) | 195. (C) | 196. (D) | 197. (C) |
| 198. (A) | 199. (A) | 200. (B) | 201. (B) | 202. (B) | 203. (D) | 204. (D) |
| 205. (A) | 206. (A) | 207. (C) | 208. (C) | 209. (A) | 210. (C) | 211. (A) |

212. (B) 213. (A) 214. (A) 215. (D)

Valence Bond Theory (VBT) : Directional properties of bonds, Overlapping of Atomic Orbitals, Types of overlapping and nature of covalent bonds

216. (A) 217. (A) 218. (B) 219. (C) 220. (C) 221. (B) 222. (C)
223. (D) 224. (B)

Hybridization

225. (A) 226. (D) 227. (B) 228. (B) 229. (A) 230. (C) 231. (D)
232. (A) 233. (C) 234. (A) 235. (C) 236. (D) 237. (C) 238. (B)
239. (D) 240. (A) 241. (C) 242. (C) 243. (C) 244. (D) 245. (B)
246. (B) 247. (B) 248. (A) 249. (C) 250. (B) 251. (A) 252. (C)
253. (B) 254. (D) 255. (C) 256. (B) 257. (A) 258. (A) 259. (C)
260. (B) 261. (B) 262. (A) 263. (A) 264. (C) 265. (B) 266. (C)
267. (C) 268. (A) 269. (D) 270. (B) 271. (C) 272. (C) 273. (C)
274. (B) 275. (C) 276. (A) 277. (C) 278. (B) 279. (A) 280. (C)
281. (A) 282. (B) 283. (C) 284. (C) 285. (C) 286. (A) 287. (A)
288. (C) 289. (A) 290. (D) 291. (D) 292. (D) 293. (C) 294. (D)
295. (A) 296. (C) 297. (D) 298. (D) 299. (A) 300. (A) 301. (C)
302. (B) 303. (C) 304. (A) 305. (A) 306. (B) 307. (B) 308. (D)
309. (D) 310. (C) 311. (A) 312. (A) 313. (A, C) 314. (A, B, D)
315. (C) 316. (A) 317. (D) 318. (A) 319. (B) 320. (C) 321. (B)
322. (A) 323. (B) 324. (A) 325. (C) 326. (A) 327. (C) 328. (C)
329. (A) 330. (D) 331. (B) 332. (B) 333. (A) 334. (A) 335. (B)
336. (C) 337. (D)

Molecular Orbital Theory : Nature of the bond, Bond-length, Magnetic nature, Bond Order Calculation, Bond Dissociation Energy

338. (A) 339. (D) 340. (A) 341. (C) 342. (B) 343. (D) 344. (C)
345. (A) 346. (B) 347. (D) 348. (D) 349. (D) 350. (D) 351. (C)
352. (A) 353. (B) 354. (B) 355. (A) 356. (A) 357. (A) 358. (C)
359. (D) 360. (D) 361. (A) 362. (A) 363. (D) 364. (B) 365. (C)
366. (B) 367. (D) 368. (B) 369. (C) 370. (A) 371. (A) 372. (D)
373. (A) 374. (B) 375. (C) 376. (C) 377. (D) 378. (D) 379. (A)
380. (B) 381. (A) 382. (C) 383. (A) 384. (B) 385. (A) 386. (A)
387. (C) 388. (A) 389. (B) 390. (B) 391. (D) 392. (C) 393. (A)
394. (A) 395. (B) 396. (D) 397. (C) 398. (D) 399. (D) 400. (A)
401. (A) 402. (C) 403. (D) 404. (A) 405. (D) 406. (A) 407. (C)
408. (B) 409. (D) 410. (C) 411. (A, B, C)

Hydrogen Bond: Types of Hydrogen Bonds

412. (C) 413. (B) 414. (B) 415. (A) 416. (C) 417. (C) 418. (C)
419. (A) 420. (C) 421. (C) 422. (C) 423. (B) 424. (D) 425. (B)
426. (A) 427. (B) 428. (D) 429. (C) 430. (A) 431. (A) 432. (C)
433. (C) 434. (C) 435. (C) 436. (A) 437. (B) 438. (A) 439. (A)

440. (A) 441. (C) 442. (D) 443. (A, C) 444. (B, D) 445. (D) 446. (A, C)
 447. (A, C, D) 448. (A, B) 449. (B, C, D) 450. (D) 451. (B) 452. (C)
 453. (D) 454. (D) 455. (B) 456. (C) 457. (D) 458. (D) 459. (B)
 460. (D) 461. (B) 462. (D) 463. (D) 464. (D) 465. (B) 466. (A)
 467. (B, D) 468. (D) 469. (C) 470. (D) 471. (D)
 472. (No d-orbital in 2nd energy level) 473. OF₄ does not have d-orbital

474. compound	Avg. FC	Avg. B.O
(i) SO ₄ ²⁻	-0.5	1.5
(ii) SO ₃ ²⁻	-0.67	1.33
(iii) ClO ₄ ⁻	-0.25	1.75
(iv) PO ₄ ³⁻	-0.75	1.25

475. (D) 476. (A) 477. (A, B, C) 478. (B) 479. (A) 480. (C, D)
 481. (D) 482. (D) 483. ($\sigma = 8, \pi = 8$) 484. (In S₂ there is no strong p-p bonding)
 485. (B) 486. (C) 487. (A, B) 488. (C) 489. (C) 490. (C) 491. (C)
 492. (D) 493. (C) 494. (B) 495. (A) 496. (D) 497. (A) 498. (C)
 499. (A, B, C) 500. (A, C) 501. (B, D) 502. (D) 503. (8) 504. (D) 505. (D)
 506. (B) 507. (C) 508. (B) 509. (A) 510. (C) 511. (C) 512. (A)
 513. (B) 514. (B) 515. (D) 516. (C) 517. (A) 518. (B) 519. (B, C, D)
 520. (A) 521. (A, B, C) 522. (5) 523. (C) 524. (C) 525. (A, C)
 526. (D) 527. (D) 528. (3) 529. (B) 530. (C) 531. (C) 532. (C, D)

V.B.T., Hybridisation & VSEPR Theory

533. (D) 534. (C) 535. (D) 536. (B) 537. (A) 538. (B) 539. (4)
 540. (D) 541. (4) 542. (D) 543. (5) 544. (B) 545. (D) 546. (C)
 547. (C) 548. (D)

Dipole Moment & Hydrogen Bonding

549. (C) 550. (D) 551. (D) 552. (D) 553. (4) 554. (B) 555. (D)
 556. (C) 557. (5)

Bent's rule & Drago's rule

558. (B) 559. (B) 560. (B)

Fajan's rule & Polarisability

561. (B) 562. (C) 563. (B)

Linear combination of atomic orbitals

564. (C) 565. (4) 566. (8) 567. (A) 568. (B) 569. (B) 570. (B)
 571. (C) 572. (5)

Molecular Orbital Theory

573. (1) 574. (A) 575. (A) 576. (B) 577. (C) 578. (4) 579. (D)

Shapes of Molecular Orbital**580. (5)****Miscellaneous****581. (C) 582. (B)****NEW QUESTIONS**

- | | | | | | | |
|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| 583. (C) | 584. (B) | 585. (B) | 586. (D) | 587. (D) | 588. (D) | 589. (D) |
| 590. (D) | 591. (C) | 592. (B) | 593. (C) | 594. (A) | 595. (A) | 596. (A) |
| 597. (B) | 598. (A) | 599. (D) | 600. (B) | 601. (A) | 602. (B) | 603. (D) |
| 604. (B) | 605. (A) | 606. (D) | 607. (C) | 608. (A) | 609. (C) | 610. (B) |
| 611. (B) | 612. (C) | 613. (B) | 614. (B) | 615. (D) | 616. (D) | 617. (B) |
| 618. (C) | 619. (A) | 620. (A) | 621. (A) | 622. (C) | 623. (D) | 624. (A) |
| 625. (D) | 626. (A) | 627. (C) | 628. (B) | 629. (C) | 630. (D) | 631. (B) |
| 632. (C) | 633. (D) | 634. (B) | 635. (B) | 636. (C) | 637. (D) | 638. (D) |
| 639. (C) | 640. (B) | 641. (A) | 642. (B) | 643. (A) | 644. (A) | 645. (C) |
| 646. (C) | 647. (C) | 648. (D) | 649. (D) | 650. (B) | 651. (D) | 652. (B) |
| 653. (D) | 654. (A) | 655. (D) | 656. (B) | 657. (B) | 658. (A) | 659. (C) |
| 660. (D) | 661. (A) | 662. (A) | 663. (C) | 664. (B) | 665. (D) | 666. (B) |
| 667. (C) | 668. (B) | 669. (C) | 670. (A) | 671. (B) | 672. (C) | 673. (C) |
| 674. (C) | 675. (A) | 676. (C) | 677. (D) | 678. (B) | 679. (A) | 680. (D) |
| 681. (C) | 682. (C) | 683. (A) | 684. (B) | 685. (D) | 686. (D) | 687. (C) |
| 688. (A) | 689. (B) | 690. (B) | 691. (A) | 692. (A) | 693. (A) | 694. (C) |
| 695. (D) | | | | | | |